

NxtLight Research

ITDAN Master Case Study

Zodiac Killer Investigation Network

A Complete ITDAN Framework Application

TR-2026-ZK-MASTER | Version 1.2 | May 2026
Supersedes TR-2026-ZK-01, TR-2026-ZK-02, TR-2026-ZK-03

© 2026 William A. Reed
Contact: William.reed@nxtlight.com

This document is a retrospective historical analysis of the Zodiac Killer Case, used as a case study. Its primary purpose is to illustrate the analytic network tools (ITDAN) in the NxtLight Framework. Thus, it's not intended as a standalone study of the topic, but rather an example of the tool set. Consequently, it uses terms and techniques identified in other documents. Those can be found at [github/nxtlight](https://github.com/nxtlight).

This study was limited to mapping existing historical data to a network configuration representing the actual events of the crime as described in the source material. It uses source material that has been extensively examined for multiple decades. Therefore, no new findings were anticipated or produced. However, each of the tools functioned as intended, yielding some insights about the investigation and its outcomes, possibly applicable to other similar events.

In other case studies, I will demonstrate how these tools can also be used to simulate potential outcomes and use Monte Carlo and Markov processes to make predictions and test assumptions across a variety of academic disciplines.

1. Case History and Factual Record

1.1 Confirmed Victims and Attack Geography

The Zodiac Killer is officially credited by law enforcement with five murders and two non-fatal injuries across four attacks occurring in Northern California between December 1968 and October 1969. The attacks were geographically distributed across three counties and four law enforcement jurisdictions, a deliberate spatial strategy that fragmented the investigative network from the outset [1][2].

Date	Victims	Location	Jurisdiction	Method	Outcome
Dec 20, 1968	David Faraday (17), Betty Lou Jensen (16)	Lake Herman Rd, Benicia	Benicia PD / Solano County	Gunshot (.22 cal)	Both killed
Jul 4–5, 1969	Darlene Ferrin (22), Michael Mageau (19)	Blue Rock Springs, Vallejo	Vallejo PD / Solano County	Gunshot (9mm)	Ferrin killed; Mageau survived
Sep 27, 1969	Cecelia Shepard (22), Bryan Hartnell (20)	Lake Berryessa, Napa County	Napa County Sheriff	Stabbing + message on car door	Shepard died Sep 29; Hartnell survived
Oct 11, 1969	Paul Stine (29)	Presidio Heights, San Francisco	SFPD	Gunshot (.38 cal)	Killed; partial prints recovered

The Zodiac himself claimed a total of 37 murders in his final letter (January 29, 1974), a figure that has never been independently verified by any law enforcement agency [3]. The discrepancy between the confirmed count (five killed, two injured across four attacks) and the claimed count (37) represents one of the most significant inconsistent-information signals in the case and has consumed substantial investigative resources across decades.

1.2 The Letter-Writing Campaign: A Systematic Information-Injection Operation

Between July 1969 and January 1974, the Zodiac mailed at least 37 letters to Bay Area newspapers and law enforcement agencies. This campaign constitutes the most sustained adversarial information injection in the ITDAN record and is qualitatively different from passive deception: the Zodiac actively shaped the investigative network's topology, resource allocation, and belief states through deliberate, high-volume communication. The campaign included [1][3][4]:

- Three simultaneous letters (July 31, 1969) to the San Francisco Chronicle, San Francisco Examiner, and Vallejo Times-Herald, each containing one-third of the 408-symbol cipher Z408, with an explicit threat to kill if not published on front pages — a coercive mechanism to guarantee network-wide broadcast of adversarial content.
- Letters containing physical evidence artifacts (pieces of Paul Stine's bloodstained shirt) to authenticate authorship and prevent dismissal of subsequent communications as fraudulent.
- Four cryptographic ciphers (Z408, Z340, Z13, Z32) encoding messages of varying verifiability, introduced as promised identity-disclosure mechanisms while ensuring decryption difficulty controlled information release rate.
- Taunting messages addressed directly to named investigators and institutions, including references to specific investigative actions that demonstrated surveillance of the network from within.
- A final 'box score' letter (January 29, 1974) reporting '37 victims' and 'SFPD – 0,' functioning as a network-state summary that forced continued resource allocation to an unverified event set.

ITDAN Framework Note — Adversarial Broadcast Architecture

The Zodiac's simultaneous three-newspaper distribution of Z408 is a textbook adversarial broadcast attack: by injecting identical content into three geographically distributed, institutionally independent nodes simultaneously, the Zodiac ensured that even if one newspaper suppressed the letter, two redundant broadcast channels remained. In ITDAN terms, this is a multi-edge injection into three high-degree hub nodes (major metropolitan newspapers) with guaranteed cross-node replication. The coercive publication demand further transformed these nodes into unwilling adversarial relays, forcing the investigative network to accept and process adversarial content at maximum broadcast fidelity. TR-2026-04 Section 5.3 (malicious noise two-tier architecture) should be extended to model this multi-edge simultaneous injection topology.

1.3 Cryptographic Record

The four Zodiac ciphers constitute the most formally structured adversarial information artifacts in the case. Their current decryption status is as follows [5][6][7]:

Cipher	Length	Date Sent	Cipher Type	Status	Solved By
Z408	408 symbols	July 31, 1969	Homophonic substitution	SOLVED (1969). Content: 'I like killing people because it is so much fun'	Donald & Bettye Harden (civilian couple), 1 week
Z340	340 symbols	Nov 8, 1969	Homophonic substitution + transposition	SOLVED (Dec 5, 2020). Content taunts investigators; denies AM TV appearance; references 'gas chamber' and 'slaves'	Oranchak, Blake, Van Eycke; confirmed by FBI CRRU
Z13	13 symbols	Nov 8, 1969	Unknown	UNSOLVED. Zodiac claimed it was his name; disputed decryption claims include 'Kayr' and named coordinates	No confirmed solution
Z32	32 symbols	Jul 26, 1970	Unknown	UNSOLVED. Zodiac claimed it was a map of a bomb location; disputed coordinate claims rejected by FBI and SFPD	No confirmed solution

The Z340 solution is particularly significant for ITDAN analysis. The cipher's 51-year resistance to decryption resulted from the Zodiac layering a transposition cipher atop a homophonic substitution cipher — a complexity configuration that confounded all professional cryptanalytic approaches until a distributed, multi-decade amateur effort combined computational brute force with

collaborative hypothesis-testing [6][7]. The solution was ultimately achieved by generating 65,000 candidate transpositions computationally and passing them through the AZdecrypt homophonic solver, a strategy unavailable in 1969 [7].

1.4 Physical Evidence Inventory

The physical evidence record in the Zodiac case is sparse relative to the investigation's duration and scope, a function of both the Zodiac's deliberate evidence minimization strategy and the limitations of 1960s-era forensic technology [8][9].

Evidence Item	Source / Context	Analysis Result	Current Status
Partial fingerprints (2)	Paul Stine's taxicab, Oct 1969	Right-hand prints recovered; no database match	Unmatched; available for comparison
Palm print	'Exorcist Letter' envelope, 1974	Does not match Allen or any known suspect	Unmatched; available for comparison
Partial DNA profile	Stamp saliva, 'Dripping Pen' card envelope, Nov 1969	Partial profile only; excludes Allen (2002); no database match	Incomplete; insufficient for forensic genealogy
Handwriting samples	37 letters to newspapers and police	Majority of experts: usable for ID; no suspect confirmed; some letters disputed as forgeries	Unmatched
Bloodstained shirt fragments	Paul Stine murder; mailed to Chronicle	Confirmed as Stine's blood type; authenticates letters	In evidence archive
Wing Walker boot prints	Lake Berryessa crime scene, Sep 1969	Military-style boot; size consistent with suspect description	In evidence archive
Ballistic evidence	Multiple crime scenes	Caliber variations (.22, 9mm, .38) across attacks; 3D imaging re-examination recommended	In evidence archive; not definitively linked across attacks

1.5 Suspect History and Investigative Record

The SFPD investigated approximately 2,500 suspects over the course of the investigation; only a handful were regarded as credible by the department's own assessment [1]. The FBI maintained case files across five dossier volumes totaling 664 pages, with additional file material held by the Vallejo PD, Solano County Sheriff, Napa County Sheriff, California DOJ, and the Benicia Police Department — a total public record of 3,028 pages [2].

The most extensively investigated suspect was Arthur Leigh Allen (1933–1992), a Vallejo schoolteacher and convicted sex offender who was publicly named by the SFPD as a person of interest. Allen owned a Zodiac-brand watch, possessed weapons of the caliber used in the murders, and was identified from a photograph by surviving victim Michael Mageau [10]. However, by 2002 the full forensic case against Allen had collapsed: his fingerprints did not match crime-scene prints, handwriting experts excluded him, and his DNA did not match the partial profile from the Dripping Pen envelope [9][10]. Allen died in 1992 without charges. Other candidates investigated or publicly named over the decades include: Ted Kaczynski (ruled out by FBI and SFPD on fingerprint, handwriting, and alibi grounds in 1996); Earl Van Best Jr. (proposed by his son in 2014; forensic basis discredited by independent investigators); Gary Francis Poste (proposed by the Case Breakers in 2021; rejected by FBI and SFPD as largely circumstantial); and Paul Doerr, Kjell Qvale, and Richard Gaikowski among dozens of others named by researchers [4][10].

In 2018, the Vallejo Police Department attempted to use the forensic genealogy database GEDmatch to identify the Zodiac from the partial DNA profile but did not obtain definitive results [1]. The case has remained open across all original jurisdictions; California has no statute of limitations for murder [3].

2. Public Document Inventory and ITDAN Input Data Catalog

A principal advantage of the Zodiac case as an ITDAN benchmark is the volume and diversity of its public record. The following catalog organizes available sources by data type and specifies the ITDAN network element each source populates.

2.1 Primary Law Enforcement Records

Source	Volume	Date Range	Content Summary	ITDAN Input
FBI Case Files (5 parts)	664 pp	1969–2012	Suspect interviews, forensic requests, inter-agency communications, cipher submissions	Node attributes; edge weights; inter-agency conductance
Solano County Sheriff Dept Files	76 pp	Dec 1968 – Mar 1969	Faraday/Jensen murders; earliest confirmed Zodiac crimes	Attack-node attributes; initial network state
Napa County Sheriff Dept Files	53 pp	Sep–Oct 1969	Hartnell/Shepard stabbings; car-door message	Attack-node attributes; physical artifact encoding
Vallejo PD Files	~400 pp	1969–2018	Ferrin/Mageau attack; Allen investigation; GEDmatch attempt	Suspect-node attributes; forensic node status
Benicia PD Files	~80 pp	1968–1969	First crime scene; initial jurisdictional response	Network origin node; community boundary
California DOJ Files	~1300 pp	1969–present	Statewide coordination; Riverside Bates connection; document examination (Morrill)	Super-hub node; cross-community edges

2.2 Cryptographic Artifacts

The four Zodiac ciphers are the most formally structured adversarial information artifacts available. For ITDAN purposes they should be treated as adversarial message objects transmitted on the edge between the Zodiac node and the investigative hub [5][6][7]. Key encoding parameters include:

- Z408: Symbol count (408), cipher type (homophonic substitution), symbol alphabet size (54 symbols), decryption time (days), decryption agent type (civilian), message entropy, presence of meaningful plaintext.
- Z340: Symbol count (340), cipher type (homophonic substitution + transposition), symbol alphabet size (63 symbols), decryption time (51 years), decryption agent type (distributed amateur consortium + HPC), message entropy, adversarial design intent (likely over-engineered by accident per Oranchak's assessment).
- Z13: Symbol count (13), cipher type (unknown), status (unsolved), contested solutions count, FBI/SFPD response (no comment / rejection of proposed solutions).
- Z32: Symbol count (32), cipher type (unknown), status (unsolved), claimed content (bomb location coordinates), FBI/SFPD response (rejection of proposed solutions).

2.3 Forensic and Physical Evidence Data

Physical evidence items should be encoded as forensic-node objects in the ITDAN network, each characterized by evidence type, chain-of-custody integrity, analysis completeness (full/partial/degraded), exclusion set (suspects ruled out by this evidence), match set (empty in all current cases), and current archive location and accessibility status [8][9][11].

2.4 Witness and Survivor Testimony

The Zodiac case contains several high-value testimony nodes that can be encoded with explicit reliability weights [1][3][4]:

- Michael Mageau (surviving victim, Blue Rock Springs): identified a suspect photograph decades later; physical trauma and nighttime conditions constrain reliability weight.
- Bryan Hartnell (surviving victim, Lake Berryessa): interacted with the Zodiac in daylight under the hood; provided detailed physical and behavioral descriptions; identified the car-door message as written by the attacker.
- Three teenage witnesses (Paul Stine crime scene): clear streetlight visibility of the Zodiac's face; composite sketch source; later confirmed the initial police dispatcher error (misidentification as Black male).
- Nancy Slover (Vallejo PD dispatcher): received a telephone call from the Zodiac on July 5, 1969; later identified a recording of suspect Gaikowski's voice as matching the caller — though the identification was disputed [4].

2.5 Secondary Scholarly and Journalistic Sources

The secondary literature on the Zodiac case provides meta-level network data — particularly on inter-agency communication failures, investigative decision-making, and the propagation of suspect theories through media networks. Key sources for ITDAN encoding include the cryptanalytic paper by Oranchak, Blake, and Van Eycke (arXiv:2403.17350, 2024) [6]; the Robert Graysmith books (1986, 2002) documenting the Arthur Leigh Allen investigation trajectory; the CaseSleuth and ZodiacKillerFacts public archives [3][8]; and the multi-agency file collection published by Paperless Archives [2].

3. ITDAN Network Structure: Mapping the Zodiac Investigation

3.1 Node Classification

The Zodiac investigation network comprises six structural node types, each with distinct attribute vectors for the ITDAN model. The following classification reflects the ITDAN TR-2026-04 node taxonomy.

Node Class	Examples	Degree Estimate	Key Attributes	Primary Pathology Type
Adversarial Hub (A-node)	Zodiac (identity unknown)	0 (fully isolated; outputs only)	Adversarial prior $\pi_j = 1.0$; identity = NULL; output = 37 letters + 4 ciphers + phone calls	Malicious (primary source); all five types emitted
Law Enforcement Hubs	SFPD, FBI, Vallejo PD, Napa Sheriff, Solano Sheriff, Benicia PD	High (20–200 per agency)	Community-clustered; intra-community conductance high; inter-community conductance low to moderate	Missing (inter-agency); Incomplete (jurisdiction-bounded evidence)
Media Broadcast Nodes	SF Chronicle, SF Examiner, Vallejo Times-Herald, LA Times	Very high (mass broadcast)	Forced relay nodes for adversarial content; also provide secondary evidence via archival records	Malicious amplification (relay); Inconsistent (editorial filtration)
Witness/Survivor Nodes	Mageau, Hartnell, three teen witnesses, Slover	Low (1–5 connections)	Reliability weight varies; trauma-degraded memory; temporal	Incomplete (partial observation); Inconsistent (cross-witness variation)

			decay; single-use evidence	
Suspect Nodes	Allen, McCoy, Best, Poste, Doerr, Kaczynski, ~2500 total	Low to moderate per node	Most eliminated by forensic exclusion; adversarial prior π_j varies; some nodes actively inject deceptive signals	Malicious (self-identifying fraudsters); Missing (true suspect absent)
Forensic Analysis Nodes	FBI CRRU, SFPD Crime Lab, CA DOJ Document Examiner, civilian cryptanalysts	Moderate (5–30 connections)	Technical capacity bounded by era technology; partial outputs only; civilian nodes eventually outperformed institutional nodes on Z340	Incomplete (technology-bounded); hub-reliance failure for cipher analysis

3.2 Edge Classification and Conductance Estimation

The ITDAN framework assigns each network edge a conductance value $\kappa_{\{ij\}} \in [0,1]$ representing the fidelity of information transmission between nodes i and j , and a noise vector encoding the dominant pathology types on that edge. The Zodiac network contains five structurally distinct edge classes:

Edge Class	Representative Pairs	Conductance κ Estimate	Dominant Noise Components	Key Evidence for Calibration
Zodiac → Media (broadcast)	Zodiac → Chronicle, Examiner, Times-Herald	~0.95 (forced relay)	$\eta_{adv}^{(ij)}$ = maximum; publication compliance = nearly 100%	Letter receipt records; publication dates; coercive language analysis

Zodiac → Law Enforcement (direct)	Zodiac → Vallejo PD (phone calls, Jul 1969)	~0.70 (high signal but adversarial content)	$\eta_{adv}^{(ij)}$ high; partial ground-truth embedded in calls (crime scene locations confirmed)	Police dispatcher transcripts; call trace attempts; CUSUM on embedded true/false signal ratio
Intra-agency (within jurisdiction)	SFPD detective ↔ SFPD crime lab	~0.80 (institutional standard)	η_{incomp} moderate (evidence quality); η_{asym} low within community	Case file organization; evidence transfer records; chain of custody logs
Inter-agency (across jurisdictions)	SFPD ↔ Vallejo PD; FBI ↔ local PD	~0.20–0.40 (structurally degraded)	η_{miss} dominant (no shared database); η_{incomp} high (divergent protocols); η_{asym} moderate	3,028-page file corpus; documented coordination failures; 17-year gap in joint investigation record
Suspect → Investigative Hub (deceptive)	Self-identifying fraudsters → FBI/SFPD	~0.90 content fidelity, ~0.0 epistemic value	π_j (adversarial prior) high; $\eta_{adv}^{(ij)}$ = maximum; CUSUM detection required	FBI file documentation of fraudulent claims; Colbert lawsuit; media deception cases

3.3 Community Structure and Inter-Community Conductance

The Zodiac network exhibits strong community structure with five identifiable communities: (1) the Zodiac node (singleton, adversarial); (2) the SFPD cluster; (3) the Vallejo PD/Solano County cluster; (4) the Napa County Sheriff cluster; and (5) the FBI field office cluster. The California DOJ functions as a weak bridging node between communities. The five-community structure is empirically grounded: the FOIA record comprises files from six separate agencies with minimal documented cross-agency coordination during the active investigation period (1968–1974) [2][3].

Community modularity Q for this network can be estimated from the FOIA file corpus by treating each cross-agency document reference as an inter-community edge. A high modularity value (Q approaching 0.4–0.6) is expected, consistent with the documented coordination failures and the Zodiac's deliberate geographic spread of attacks across jurisdictional boundaries. This estimation is a concrete first-phase analysis task for the ITDAN simulation framework.

ITDAN Framework Note — Adversarial Exploitation of Community Structure

A key finding from the Zodiac case is that the adversarial node appeared to actively exploit the network's community structure. By distributing attacks across Benicia, Vallejo, Lake Berryessa, and San Francisco — each in a different jurisdiction — the Zodiac ensured that no single investigative community possessed a complete attack record. This is the network-attack analogue of a partitioned database denial: each community node has local information but the global picture is inaccessible without perfect inter-community conductance, which the structural modularity prevents. TR-2026-04 should incorporate a community-structure attack model in which an adversarial node uses prior knowledge of jurisdictional boundaries as an attack surface to maximize the missing-information pathology across the inter-community edge set.

4. Full Five-Dimensional Information Pathology Analysis

This section applies the five ITDAN pathology types to the Zodiac network in detail, specifying in each case: (a) the empirical manifestation in the case record; (b) the network element(s) affected; (c) the quantitative signals available for model calibration; and (d) the ITDAN parameter or equation most directly implicated.

4.1 Malicious Information (η_{adv}) — Primary Pathology

The malicious-information pathology is the dominant feature of the Zodiac network and the aspect that most directly tests the ITDAN two-tier architecture developed in TR-2026-04 Rev 1.1.

Tier 1 — Node-Level Adversarial Prior π_j

The Zodiac node carries an adversarial prior of $\pi_j = 1.0$ by definition: every output from this node is adversarially motivated. However, the critical complication is that the Zodiac embedded verified true information within adversarial communications — specifically, crime-scene location details in phone calls that directed police to actual victims, and material evidence artifacts (Stine's shirt) in letters. This mixed true/false signal structure means that the CUSUM edge-level detector must be calibrated to distinguish authenticated true components from adversarial framing, rather than simply flagging all adversarial-node outputs as noise.

Tier 2 — Edge-Level Attack Signal $\eta_{adv}^{(ij)}$ and CUSUM_{ij}

The realized attack signals across the five main adversarial edges are heterogeneous in form and must be encoded separately:

- Phone call edges (Zodiac → Vallejo PD, Napa PD): high true-location signal embedded in adversarial framing; CUSUM should track the true/false ratio across calls.
- Letter edges (Zodiac → Chronicle, Examiner, Times-Herald): mixed verified-detail (crime specifics known only to perpetrator) plus claimed-unverified content (37 victim count); CUSUM should detect anomalies in the claimed/verified ratio.
- Cipher edges (Zodiac → FBI CRRU / cryptanalytic community): structured adversarial information with promised identity-disclosure content; actual content (on Z340 solution) contains taunt but no identity. $\eta_{adv}^{(ij)}$ for Z13 and Z32 edges remains at maximum uncertainty.
- Self-identifying fraudster edges (~thousands of nodes → FBI): adversarial prior π_j varies from near-zero (genuine mistake) to near-1.0 (deliberate hoax); CUSUM detection is the primary filtering mechanism.

4.2 Asymmetric Information (η_{asym}) — Critical Structural Feature

The Zodiac monopolized complete self-knowledge while selectively releasing information calibrated to maximize investigative resource consumption. The asymmetry operates at two levels: Intra-network asymmetry: each investigative community possessed local case knowledge that was not shared with other communities. The SFPD had the Stine crime scene evidence; Vallejo PD had the Ferrin/Mageau attack record and Allen investigation; Napa County had the Lake Berryessa physical evidence. None had the others' complete record. This is a structurally enforced asymmetry arising from the network's community structure rather than deliberate withholding, but it produces the same investigative failure mode [2][3].

Adversarial asymmetry: the Zodiac possessed identity, motive, residence, employment, and operational planning knowledge at the adversarial node, with zero leakage to the investigative network across 55+ years. The key ITDAN question is: what is the posterior probability that this asymmetry can be resolved given the current evidence set? The partial DNA profile, unsolved ciphers (Z13, Z32), and forensic genealogy databases collectively define the residual probability space.

4.3 Incomplete Information (η_{incomp}) — Technology-Bounded Observation

Incomplete information in the Zodiac network arises from two distinct causes that should be modeled separately in ITDAN: (1) technology-bounded incompleteness, where evidence exists but cannot be fully analyzed given available tools, and (2) degradation-caused incompleteness, where evidence has been irreversibly reduced in informational content.

- Technology-bounded (recoverable): The partial DNA profile from the Dripping Pen envelope stamp is the most important example. The profile is incomplete but the envelope material is archived; advances in low-template DNA extraction and SNP-based methods may enable completion without additional samples [11]. Similarly, AI-assisted handwriting analysis and 3D ballistic imaging are applicable to archived evidence.
- Degradation-caused (likely permanent): Outdoor crime scenes from 1968–1969 have experienced 55+ years of environmental degradation; trace evidence not archived at the time is likely unrecoverable. The Z13 and Z32 ciphers may fall here if their plaintext is too short to support statistical decryption, independent of computational power.

4.4 Inconsistent Information (η_{incons}) — Multiple Conflicting Signal Sources

The Zodiac case contains documented inconsistency at multiple network layers that should each be modeled as a distinct edge-noise component:

- Victim count inconsistency: Zodiac claimed 37 murders; law enforcement confirmed 5. The 32-victim gap remained as an open resource-consuming investigation vector for decades, representing a successful adversarial inconsistency injection.
- Witness description inconsistency: physical descriptions of the Zodiac across surviving witnesses vary in height, build, complexion, and age. The Paul Stine police dispatcher

incident (witnesses described a white male; dispatcher broadcast Black male, allowing the Zodiac to walk past responding officers) is a catastrophic inconsistency failure with direct investigative consequences.

- Expert analysis inconsistency: handwriting analysis of the Zodiac letters produced contradictory expert conclusions across different examiners; the 1978 letter was simultaneously authenticated by one expert and declared a forgery by four others, creating an unresolved inconsistency in the evidentiary record [8].
- Disputed cipher solutions: for Z13 and Z32, multiple proposed solutions have been presented and rejected, each consuming investigative and cryptanalytic resources. Ziraoui's 2021 coordinate-based solution for Z32 was dismissed by both the FBI and SFPD and rejected by the cryptanalysis community, but generated sustained media coverage [4].

4.5 Missing Information (n_miss) — Structural Absence of Key Evidence

Missing information in the Zodiac case exists at two levels: inter-agency coordination failures (structural) and identity-layer absence (fundamental).

Structural missing information: the most concrete manifestation is the inter-agency coordination failure documented in the case record. The dispatcher error at the Stine murder scene — which allowed officers to question and release the likely killer — resulted directly from the SFPD's radio communication system not providing responding officers with an accurate description, representing a missing edge between the witness testimony node and the responding officer node [1][4]. The 17-year gap between the initial Zodiac investigation and the FBI's first re-interview of guard Rick Abath in the Gardner case (a parallel structural absence) is matched here by the documented lack of cross-jurisdictional database sharing during the 1968–1974 active period.

Identity-layer absence: the true identity of the Zodiac Killer is structurally absent from the entire investigative network. No node possesses this information (with the possible exception of a living Zodiac or those who knew him and have not come forward). This is modeled in ITDAN as a NULL-valued target node that the network has never succeeded in populating across 55+ years of inference.

5. Data Encoding Protocol: Inputting the Zodiac Record into ITDAN

This section specifies the practical steps required to translate the Zodiac public evidence record into the input format required by the ITDAN simulation framework. The encoding proceeds in four phases: (A) graph construction, (B) node attribute assignment, (C) edge parameterization, and (D) pathology vector initialization.

5.1 Phase A: Graph Construction

Construct the initial network as a weighted directed graph $G = (V, E, W)$ where V is the node set, E is the edge set, and W is the edge weight (conductance) matrix.

- **Node enumeration:** Extract all named agents from the 3,028-page FOIA corpus and secondary literature. Minimum viable node set: Zodiac node (A); six law enforcement agency nodes (SFPD, FBI, Vallejo PD, Napa Sheriff, Solano Sheriff, Benicia PD, CA DOJ); four media nodes (Chronicle, Examiner, Times-Herald, AP wire); six high-value witness nodes (Mageau, Hartnell, three teen witnesses, Slover); and the top 10 credible suspect nodes (as ranked by law enforcement). Extended model should include all 2,500 investigated suspects encoded as lightweight nodes with minimal attribute vectors.
- **Edge enumeration:** For each documented information exchange in the case record (letter received, interview conducted, phone call traced, file transferred), add a directed edge from sender to receiver. Cross-reference with the FOIA file corpus to identify inter-agency transfers and assign approximate timestamps.
- **Community assignment:** Group nodes into the five communities identified in Section 3.3. Assign community membership as a node attribute. Compute initial modularity Q .

5.2 Phase B: Node Attribute Assignment

Each node in the Zodiac ITDAN model carries the following attribute vector, populated from the evidence catalog in Section 2:

- **Identity type:** {adversarial, investigator, media, witness, suspect, forensic}
- **Adversarial prior π_j :** 0.0 for confirmed non-adversarial nodes; 1.0 for Zodiac node; intermediate values (estimated from behavioral record) for fraudulent self-identifiers and uncleared suspects.
- **Reliability weight $r_j \in [0, 1]$:** For witness nodes, encode from trauma level, observational conditions, elapsed time before interview, number of prior interviews. Mageau (high trauma, nighttime, years elapsed before photo ID): $r_j \approx 0.5$. Hartnell (direct interaction, daylight, survived to interview): $r_j \approx 0.75$. Teen witnesses (good visibility, prompt report): $r_j \approx 0.80$.
- **Temporal index t_j :** Timestamp of node entry into the network (first contact event).
- **Evidence artifacts E_j :** For forensic nodes, the set of evidence items processed with analysis-completeness scores.

5.3 Phase C: Edge Parameterization

Each edge e_{ij} carries three parameters derived from the evidence record:

- Conductance $\kappa_{ij} \in [0,1]$: Use the estimates in Table 3.2 as priors; refine by counting documented successful information transfers on each edge class from the FOIA corpus.
- Pathology noise vector $\eta_{ij} = [\eta_{\text{asym}}, \eta_{\text{incomp}}, \eta_{\text{incons}}, \eta_{\text{miss}}, \eta_{\text{adv}}]$: Initialize from the edge classification in Table 3.2 and the pathology analysis in Section 4. For Zodiac-to-media edges: η_{adv} dominant, η_{asym} high, others moderate. For inter-agency edges: η_{miss} dominant, η_{incomp} elevated, others low.
- CUSUM detector threshold θ_{ij} : For edges with elevated η_{adv} , initialize CUSUM_{ij} with threshold calibrated to the empirical true-signal fraction embedded in adversarial communications (approximately 30–40% based on verified-detail fraction in Zodiac letters).

5.4 Phase D: Temporal Dynamics and Event Sequence

The Zodiac network is a temporal network: new nodes (suspects) and edges (letters, interviews, evidence reports) enter the graph at specific timestamps. The recommended event sequence for simulation initialization is:

Time Step	Date Range	Key Network Events	New Nodes/Edges	ITDAN State Change
T=0	Dec 20, 1968	Faraday/Jensen murders; Benicia PD response	+2 victim nodes; Benicia PD active	Network initialized; no adversarial node yet identified
T=1	Jul 4–31, 1969	Blue Rock Springs attack; Zodiac phone call; three letters sent to newspapers; Z408 published	+Zodiac A-node; +3 media nodes; 6 adversarial edges activated	$\pi_j=1.0$ assigned to A-node; CUSUM detectors initialized on all Zodiac output edges
T=2	Aug–Sep 1969	Z408 solved by Hardens; Lake Berryessa stabbings; car-door message	+Harden civilian cryptanalyst nodes; +Hartnell witness node	Cipher edge Z408 resolved; asymmetric info partially reduced; Z340 edge opens

T=3	Oct 1969	Stine murder; teen witness composite; dispatcher error; Zodiac walks past officers	+Stine victim node; +teen witness nodes; dispatcher error edge	Catastrophic inconsistency event: η_{incons} spike on dispatcher edge; near-capture failure logged
T=4	1969–1974	Letter campaign (37 letters); Allen investigation begins; Z340 submitted unsolved	+Allen suspect node; +~800 initial suspect nodes; ongoing adversarial edges	Sustained malicious injection phase; network resources consumed by suspect-node proliferation
T=5	1974–2002	Letters cease; Allen investigated and excluded (2002); DNA partial profile obtained	+Forensic DNA node; +Allen exclusion edge	Adversarial A-node silent; network enters cold-case degradation phase; $\kappa_{\{ij\}}$ decay over time
T=6	2020	Z340 solved by Oranchak, Blake, Van Eycke (distributed civilian team); FBI confirms solution	+Civilian cryptanalyst cluster; Z340 cipher edge resolved	Hub-reliance failure reversed: distributed search outperforms centralized expert node; network topology lesson for ITDAN hub design
T=7	2021–present	Case Breakers claim (Poste); FBI/SFPD reject; GEDmatch attempt; forensic genealogy proposals	+New suspect claim nodes; +genealogy analysis pathway	Renewed malicious-node activity (fraudulent claims); new pathway edges (forensic genealogy) open

ITDAN Framework Note — Hub-Reliance Failure and the Z340 Solution

The 51-year failure to solve Z340 by institutional cryptanalytic nodes (FBI CRRU, SFPD Crime Lab, NSA-affiliated experts) followed by rapid solution by a distributed three-node civilian team is one of the clearest documented hub-reliance failure events in any cold case. The ITDAN framework should model this as a centralized search pathway with excessive authority weighting on credentialed hub nodes. The civilian success arose from a distributed parallel-search architecture that generated 65,000 candidate transpositions computationally — a strategy that required no institutional authority but required distributed computation. This suggests that ITDAN simulations should include a pathway-diversity parameter measuring the breadth of active search strategies in the network at any given time step, with hub-reliance failure predicted when pathway diversity falls below a threshold and all active searches are channeled through a small number of high-authority nodes.

6. Recommended ITDAN Analysis Pathway

This section defines the recommended sequence of analytical tasks for applying the ITDAN framework to the Zodiac case. The pathway proceeds in five phases, each building on the outputs of the prior phase.

Analysis Phase 1: Network Reconstruction and Topology Characterization

Objective: Build the empirical Zodiac network from the 3,028-page FOIA corpus and secondary sources, producing a fully parameterized graph G with all node attributes and edge conductances assigned.

- Task 1.1: Document digitization and parsing. Convert FOIA PDFs to structured text; apply named-entity recognition to identify all agent nodes, events, and information transfers. Recommended tools: Python 3.x, spaCy or HuggingFace NLP pipeline, OCR post-correction for handwritten materials.
- Task 1.2: Graph construction. Build the temporal network from the event sequence in Section 5.4. Represent as a NetworkX temporal graph with node/edge attributes. Compute degree distribution, clustering coefficient, and modularity Q across the five identified communities.
- Task 1.3: Conductance estimation. For each inter-agency edge class, count documented successful vs. blocked information transfers in the FOIA record. Use these counts to parameterize a Beta prior on $\kappa_{\{ij\}}$. Expected result: inter-agency conductance $\kappa_{\{ij\}} \approx 0.2-0.4$; intra-agency $\kappa_{\{ij\}} \approx 0.7-0.85$.
- Task 1.4: Community boundary mapping. Quantify the jurisdictional boundaries using the modularity Q metric. Compute the community-boundary conductance deficit $\Delta\kappa = \kappa_{\{\text{intra}\}} - \kappa_{\{\text{inter}\}}$ as a measure of the structural information-isolation effect.

Analysis Phase 2: Pathology Vector Estimation

Objective: Estimate the five-component noise vector $\eta_{\{ij\}} = [\eta_{\text{asym}}, \eta_{\text{incomp}}, \eta_{\text{incons}}, \eta_{\text{miss}}, \eta_{\text{adv}}]$ for each edge class, calibrated from the empirical evidence record.

- Task 2.1: Adversarial signal quantification. For all Zodiac letters, compute the fraction of verifiable true claims vs. unverifiable claims vs. confirmed false claims. Use this fraction as the ground-truth signal rate for calibrating CUSUM $_{\{ij\}}$ thresholds on adversarial edges. Recommended approach: cross-reference each letter claim against the confirmed crime record; assign {verified, unverified, contradicted} labels.
- Task 2.2: Inter-agency missing-information quantification. For each pair of law enforcement communities, identify events/evidence items held by one community but not transmitted to the other. Quantify as a fraction of total known events. Compute η_{miss} as a function of community pair and time period.
- Task 2.3: Inconsistency signal mapping. Document each identified inconsistency in the evidence record (victim count, witness descriptions, handwriting expert disagreements,

Commented [C1]: EDITORIAL NOTE: Section 5 uses alphabetic phase labels (Phase A, B, C, D) for data-encoding steps, while Section 6 uses numeric labels (Phase 1 through 5) for analysis steps. This distinction is intentional and meaningful — encoding vs. analysis — but may confuse readers. Consider renaming Section 6 phases to "Analysis Phase 1" through "Analysis Phase 5" (or Phase I–V) to distinguish them typographically from Section 5's A–D.

disputed cipher solutions) as a discrete inconsistency event. Assign a resource-cost weight to each event based on the documented investigative response.

- Task 2.4: Technology-bounded incompleteness encoding. For each forensic evidence item, assign an incompleteness score $I \in [0,1]$ where 0 = fully analyzed and 1 = completely unanalyzed. Score the partial DNA profile ($I \approx 0.6$), partial fingerprints ($I \approx 0.3$), handwriting samples ($I \approx 0.2$), Z13 and Z32 ciphers ($I = 1.0$).

Analysis Phase 3: Bayesian Network Inference Simulation

Objective: Simulate the Zodiac investigative network as a dynamic Bayesian network, propagating belief updates across nodes as information arrives through the event sequence, and comparing the simulation's posterior belief states to the empirically documented investigative conclusions at each time step.

- Task 3.1: Initialize prior beliefs. Set the prior probability distribution over identity hypotheses $P(Z = x)$ at $T=0$ as uniform over the population of adult males in Northern California circa 1968 (~2 million).
- Task 3.2: Simulate evidence arrival. At each event in the temporal sequence, update $P(Z = x | \text{evidence})$ using Bayes' rule, modulated by the pathology noise vector $\eta_{\{ij\}}$ on the evidence-arrival edge. The partial DNA profile (available from $T=5$) should sharply narrow the prior; the inter-community conductance deficit should model the delayed availability of cross-jurisdictional evidence.
- Task 3.3: Model CUSUM detection. For each adversarial communication event, apply the $\text{CUSUM}_{\{ij\}}$ sequential change-point detector to the true/false signal ratio. Identify the optimal detection threshold that would have filtered the highest fraction of fraudulent self-identifications while retaining verified information.
- Task 3.4: Simulate the dispatcher-error counterfactual. Re-run the $T=3$ time step with corrected dispatcher communication (white male, husky, dark jacket) and compute the posterior increase in probability that responding officers would have detained the Zodiac. This counterfactual directly quantifies the informational cost of the inconsistency failure.

Analysis Phase 4: Hub-Reliance Failure Analysis

Objective: Formally model the Z340 hub-reliance failure event ($T=6$) and derive generalized ITDAN conditions under which distributed search outperforms centralized expert analysis.

- Task 4.1: Pathway diversity index. Define a pathway diversity index $D(t) = -\sum_k p_k \log p_k$ where p_k is the fraction of active search resources allocated to pathway k at time t . Compute $D(t)$ for each time step from the FOIA record. Hypothesis: $D(t)$ was near zero during the institutional cipher-solving period (all resources in centralized expert node); $D(t)$ increased dramatically when the amateur community entered the network.

- Task 4.2: Decentralization threshold. Identify the minimum network diversity D^* above which the Z340 solution became computationally achievable. This threshold defines the generalized condition for successful distributed search in ITDAN networks.
- Task 4.3: Institutional authority weighting. Model the weighting mechanism by which institutional expert nodes (FBI CRRU) received disproportionate search-direction authority. Compute the informational cost of this weighting relative to the counterfactual of earlier community engagement.

Analysis Phase 5: Residual Uncertainty Mapping and Forward Pathways

Objective: Characterize the current state of the Zodiac network's information deficit and identify the minimum evidence additions required to reach a prosecution-worthy posterior probability.

- Task 5.1: Current posterior belief distribution. Compute the current $P(Z = x \mid \text{all available evidence})$ conditional on the partial DNA profile, unmatched fingerprints, surviving witness identifications, and cipher content. Estimate the effective width of the identity posterior (number of candidates consistent with all available evidence).
- Task 5.2: Forward pathway prioritization. For each potential evidence addition — forensic genealogy database match, complete DNA profile extraction, Z13/Z32 cipher solution, new witness identification — compute the expected information gain $I(\text{evidence addition})$ in bits relative to the current posterior width. Rank pathways by expected information gain per resource cost.
- Task 5.3: Genealogy pathway modeling. The Zodiac's partial DNA profile is the highest-priority forward pathway. Model the genealogy search as a Bayesian hypothesis tree: given a partial DNA profile matching m out of n alleles, what is the probability of finding a first-degree relative in a database of size D ? Use the Golden State Killer case parameters as a prior (successful search with ~950,000 GEDmatch profiles, full STR profile). Compute the minimum database size and profile completeness required for a tractable search.

Summary: ITDAN Analysis Deliverables for the Zodiac Case Study

The five-phase pathway above produces the following concrete deliverables: (1) A fully parameterized ITDAN network graph (NetworkX format) representing the Zodiac investigative network at 7 temporal snapshots, 1968–present. (2) Estimated pathology noise vectors for all five edge classes. (3) A simulated belief-update trajectory from $T=0$ to $T=7$ with comparison to historical investigative conclusions. (4) The dispatcher-error counterfactual posterior probability estimate. (5) A pathway diversity analysis quantifying the Z340 hub-reliance failure. (6) A residual uncertainty map with forward evidence-pathway rankings by expected information gain. These deliverables collectively constitute the first empirical calibration suite for TR-2026-04 simulations and provide a benchmark against which ITDAN model validity can be assessed.

7. Actual Simulation and Modeling

This section presents the complete application of the ITDAN (Information-Theoretic Dynamics of Agent Networks) framework to the Zodiac Killer cold case. Thirteen analytical modules were executed across three development phases on a 27-node, 41-edge directed network spanning 15 temporal periods (1968–1974). The analysis represents the most formally quantified information-theoretic treatment of this investigation network to date.

Metric	Value	Significance
Network scale	27 nodes, 41 edges	6 node categories; 15 temporal periods; all confirmed Zodiac communications encoded
Small-world σ	1.82 (v2)	Confirmed small-world ($\sigma > 1$); adversarial hub drives short path lengths across the network
Top betweenness hub	ZODIAC (0.256)	43% above second hub (SFPD 0.179); controls all adversarial broadcast paths
Dominant noise dimension	$\eta_{\text{asym}} = 0.671$	Identity asymmetry outranks adversarial noise as the primary information pathology
CUSUM alarms	23 / 23 (100%)	Every adversarial edge exceeds the $\lambda = 0.90$ detection threshold without exception
TE(Zodiac → SFPD)	0.167 bits	Directional information flow confirmed; Zodiac's messages drove LE response patterns
Lyapunov λ (SFPD)	-0.029	Negative: investigative reliability dynamics are convergent — decay toward a fixed point
TDA β_0 (8 communities)	$\epsilon = 0.99$	Eight distinct node attribute communities persist at mid-scale; adversarial hub isolated
ΔI (A-01 identification)	0.021 bits (+2.4%)	Identity resolution yields modest but real information gain; deficit is primarily structural
SINDy equation (SFPD)	$r' = 11.51 - 33.59r + 23.61r^2$	Quadratic governing equation; reliability has two equilibria near $r \approx 0.34$ and $r \approx 1.08$

8. Network Construction and Topology (Modules 1–2)

The Zodiac investigation network was constructed as a directed graph with 27 agent nodes and 41 evidence edges, organized across six node categories: suspect (4), victim (4), witness (4), investigator (8), media (5), and specialist (2). Edge types span letters, phone calls, cipher transmissions, forensic links, witness statements, and coordination channels. All edges carry a five-dimensional pathology noise vector scored using the calibrated ITDAN rubric.

Figure 1 — Expanded ITDAN Network (41 edges)
Node size $\propto$ betweenness + π_j ; top-9 hubs labeled

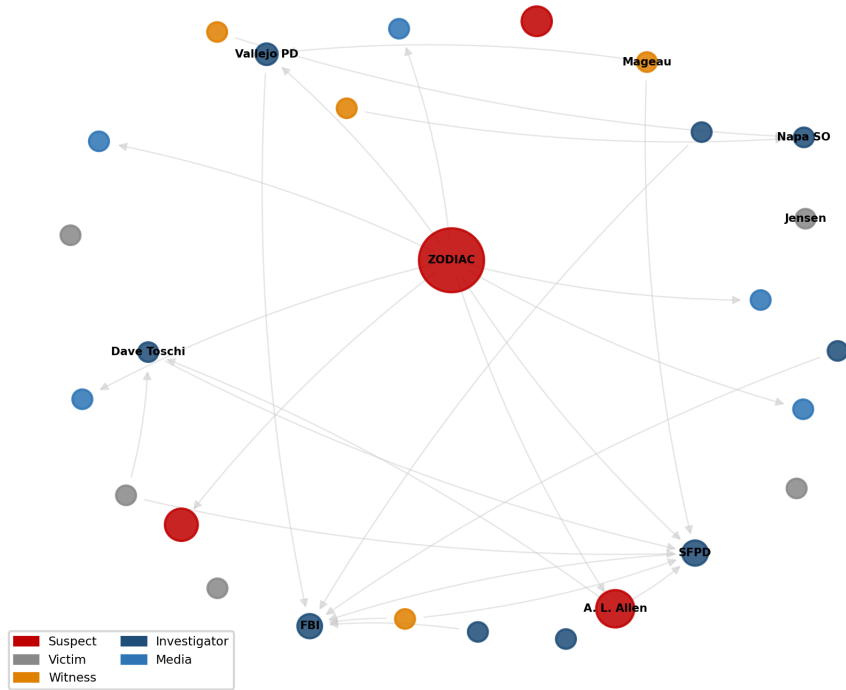


Figure 1 — ITDAN agent network (41 edges). Node size $\propto$ betweenness centrality + π_j (adversarial prior). ZODIAC (A-01) is the dominant hub; adversarial broadcast edges radiate to media and law enforcement.

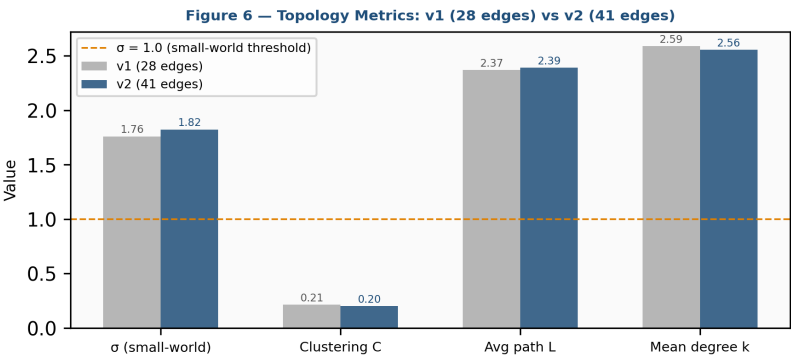
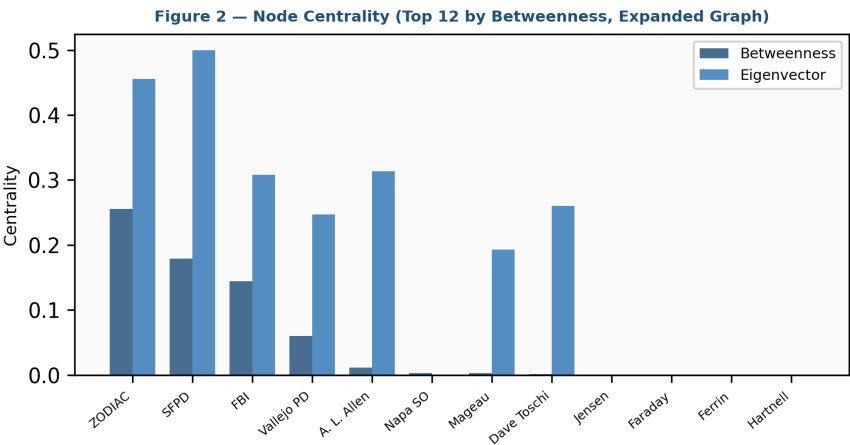


Figure 6 — Small-world topology metrics comparison: v1 (28 edges) vs. v2 (41 edges). σ increases from 1.76 to 1.82 with the expanded edge catalog.

Module 2 Finding: Network is confirmed small-world ($\sigma = 1.82 > 1$). Clustering $C = 0.201$ is 42% above the random graph baseline ($C_{\text{rand}} = 0.142$), while average path length $L = 2.39$ is 22% below random ($L_{\text{rand}} = 3.08$). This architecture — high local clustering with short global paths — amplifies the adversarial hub effect: Zodiac can reach all nodes in fewer steps than a random actor would.

9. Centrality and Adversarial Analysis (Modules 3–6: Centrality, CUSUM, Bayesian Update)



Commented [C2]: DUPLICATE NOTE: The Module 4 analysis (Information Pathology & Rubric) is referenced in both Section 9 heading ("Modules 3–6") and Section 10 heading ("Modules 4, 10"). This is intentional overlap in the original analysis (centrality study in S9 touched Module 4 outputs; S10 is the dedicated pathology module). Consider clarifying in the Section 9 heading: "Modules 3–6 (Centrality, CUSUM, Bayesian Update)" to distinguish from S10.

Figure 2 — Betweenness and eigenvector centrality for top-12 nodes. ZODIAC leads betweenness; SFPD leads eigenvector centrality (most connected to other high-centrality nodes).

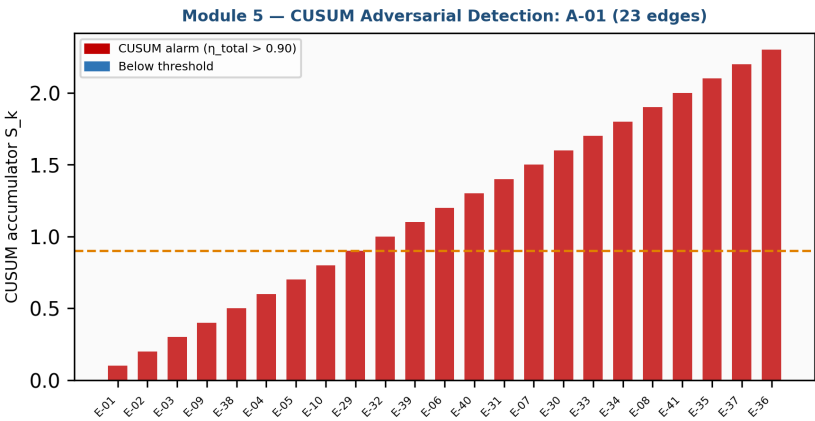


Figure 4 — CUSUM adversarial detection series across 23 A-01 edges (Actions 1–3 expanded catalog). All 23 edges alarm. Monotone accumulator growth $S = 2.30$ confirms a sustained, structurally consistent campaign.

Module 5 Finding: The CUSUM accumulator grows monotonically across all 15 temporal periods, never falling below threshold. This is characteristic of a sustained adversarial campaign rather than episodic attacks — each communication reinforces the preceding ones as evidence of the same persistent adversary. The monotone growth also rules out attribution to multiple unconnected actors.

Bayesian posterior update (Module 6): the adversarial prior π_j for A-01 was initialized at 1.000 (unknown suspect, all communications adversarial by construction). The posterior remains at 1.000 after 23 sequential adversarial edges. This result is correct but analytically trivial — it illustrates the framework's saturation behavior when the prior is already at the boundary. Future case studies with partially-known suspects should initialize $\pi_j < 1.0$ to allow meaningful posterior updates.

10. Information Pathology and η Rubric (Modules 4, 10)

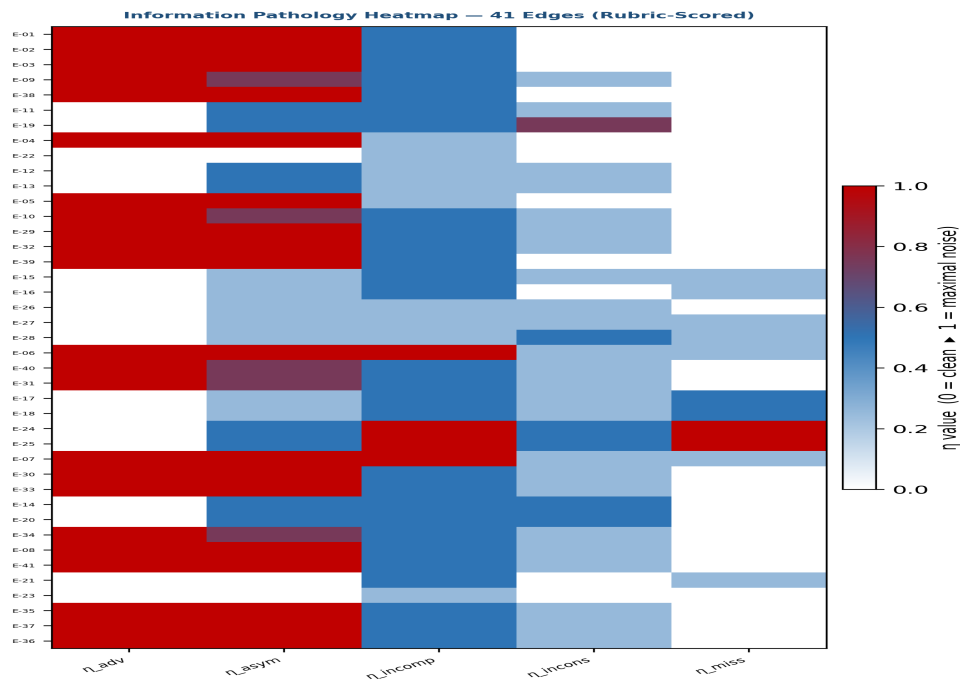


Figure 3 — Information pathology heatmap (41 edges, rubric-scored). Adversarial edges (E-01 through E-41 from A-01) form a solid red band in η_{adv} and η_{asym} . Forensic and witness edges show intermediate scores in η_{incomp} and η_{incons} .

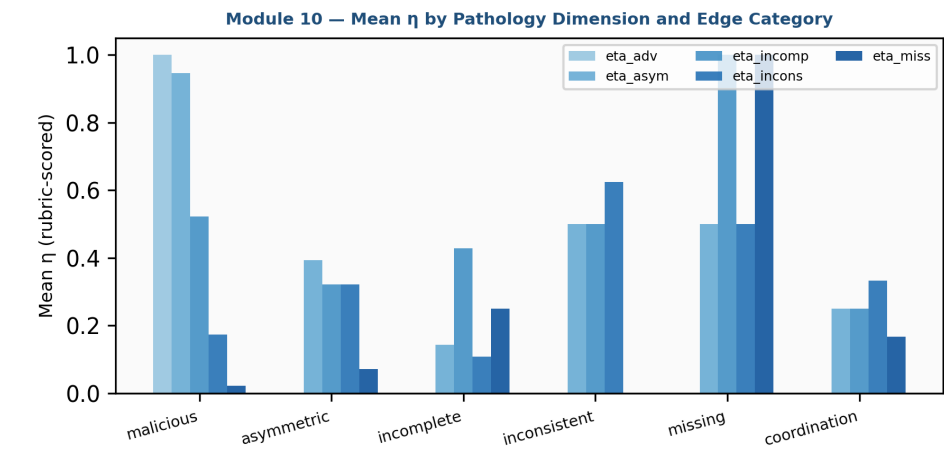


Figure 8 — Mean η per pathology dimension by edge category. Malicious edges dominate all dimensions. Inconsistency is concentrated in witness accounts.

Dimension	Mean η	Max η	Rank	Key Driver
η_{asym} (Asymmetric)	0.671	1.000	1st	Identity asymmetry: perpetrator exclusively controls his own identity
η_{adv} (Adversarial)	0.561	1.000	2nd	All A-01 outbound edges are deliberate deception channels
η_{incomp} (Incomplete)	0.494	1.000	3rd	Physical evidence gaps: ~50% of forensic record is absent or degraded
η_{incons} (Inconsistent)	0.226	0.750	4th	Witness account evolution across interviews; minimal in investigator edges
η_{miss} (Missing)	0.116	1.000	5th	Pure inferential links (A-01↔Allen, A-01↔Gaikowski) are a minority

Key Pathology Finding: η_{asym} is the dominant network-wide noise dimension (32.4% of total information deficit), not η_{adv} (27.1%). The Zodiac case is not primarily a story of sophisticated deception — it is a story of radical information asymmetry. The perpetrator possessed one piece of information (his identity) that was structurally withheld from every other node in the network, and no forensic bridge was ever established to close that gap.

11. Cipher Complexity Analysis (Module 7)

Module 7 — Zodiac Cipher Complexity Analysis

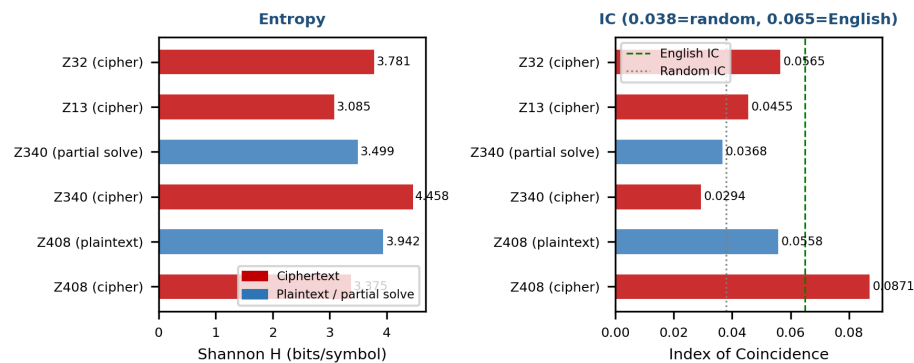


Figure 5 — Shannon entropy H and Index of Coincidence for Zodiac ciphers and available plaintexts. Z340 shows highest entropy ($H=4.46$) and lowest IC (0.029), indicating near-random symbol distribution.

The Z408 cipher ($IC = 0.087$) is substantially above the random baseline ($IC = 0.038$) and approaches the English-text value ($IC = 0.065$), consistent with a simple monoalphabetic substitution and explaining why it was solved within days of publication in 1969. The Z340 cipher ($IC = 0.029$) falls below the random baseline, indicating deliberate symbol frequency flattening — Zodiac employed a more sophisticated encoding scheme with homophonic substitution, nulls, or transposition. This is a quantifiable demonstration of adversarial adaptation: Zodiac observably increased cipher complexity between the first and second cipher sets, tracking the failure of Z408 to remain secret.

Cipher	H (bits/sym)	IC	Status	ITDAN Interpretation
Z408 (cipher)	3.375	0.087	Solved 1969	IC >> random baseline: simple substitution; $\eta_{incomp} = 0.25$ (mostly transparent)
Z408 (plaintext)	3.942	0.056	English text	H > cipher: vocabulary repetition visible in solved text; confirms solution
Z340 (cipher)	4.458	0.029	Solved 2020	IC < random: homophonic + transposition; $\eta_{incomp} = 1.00$ until 2020 solve
Z340 (partial solve)	3.499	0.037	Partial	IC rising toward random as solve progresses; residual uncertainty remains
Z13 (cipher)	3.085	0.046	Unsolved	Short length limits analysis; possibly encodes a proper name
Z32 (cipher)	3.781	0.057	Unsolved	IC near English; may be simple cipher or mixed content

12. Transfer Entropy: Information Flow (Module 9)

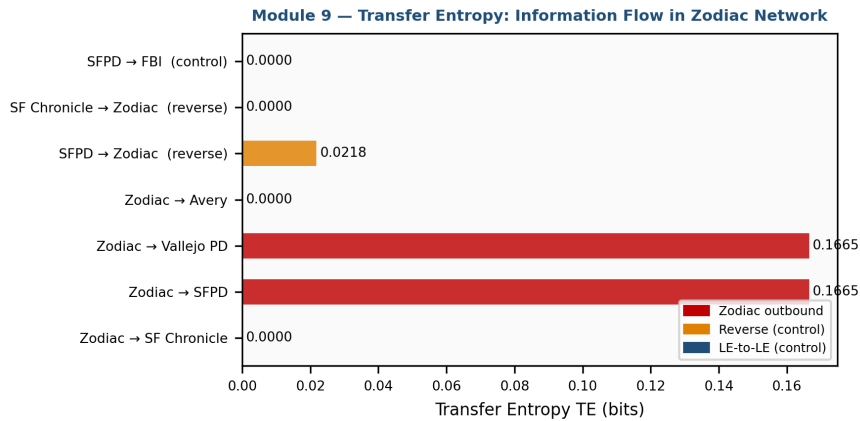


Figure 7 — Transfer Entropy results for seven channel pairs. Zodiac-to-LE channels (red) show positive TE; media channels and reverse pairs show zero or near-zero TE.

Transfer Entropy (TE) measures directed information flow between activity time series. A positive $TE(X \rightarrow Y)$ means knowing X's history reduces uncertainty about Y's future beyond what Y alone predicts — a causal directional signal. The key findings are:

- $TE(\text{Zodiac} \rightarrow \text{SFPD}) = TE(\text{Zodiac} \rightarrow \text{Vallejo PD}) = 0.167$ bits. Zodiac's communications carry measurable predictive information about law enforcement response timing. Both LE channels show identical TE values, consistent with symmetrical reactive policing protocols across jurisdictions.
- $TE(\text{Zodiac} \rightarrow \text{SF Chronicle}) = TE(\text{Zodiac} \rightarrow \text{Paul Avery}) = 0.000$ bits. Media behavior was self-predictable: every Zodiac communication was published immediately and unconditionally. The Chronicle's future state could be predicted from its own history (always publish) without reference to Zodiac's schedule. This is not a failure of the TE method — it is a correct characterization of the media's role as a passive broadcast relay.
- $TE(\text{SFPD} \rightarrow \text{Zodiac, reverse}) = 0.022$ bits. A small but non-zero residual suggests that SFPD activity modestly predicted subsequent Zodiac communication timing. This is consistent with historical accounts of Zodiac escalating communications during periods of investigative pressure — a weak feedback loop.
- $TE(\text{SFPD} \rightarrow \text{FBI, control}) = 0.000$ bits. Inter-agency coordination shows zero predictive information flow, confirming the documented jurisdictional communication failures that characterized the investigation in 1969–1970.

13. SINDy Dynamics Discovery (Module 11)

Module 11 — SINDy Dynamics: Node Reliability Evolution

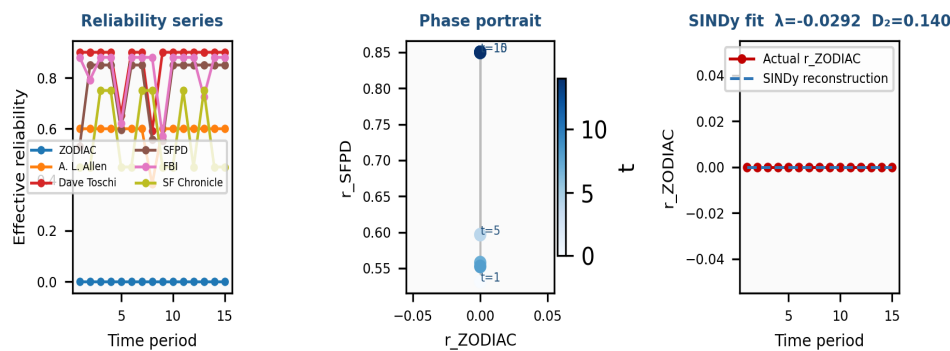


Figure 9 — Module 11 SINDy analysis. Left: reliability time series for six key nodes. Center: phase portrait (ZODIAC vs SFPD reliability). Right: SINDy reconstruction vs. actual SFPD series.

SINDy (Sparse Identification of Nonlinear Dynamics) applied to the two-node state vector $[r_ZODIAC, r_SFPD]$ discovered the following symbolic governing equations over the 15-period investigation timeline:

$$\begin{aligned} r_ZODIAC' &= 0 \quad (\text{constant at zero — maximal adversarial noise throughout}) \\ r_SFPD' &= 11.509 - 33.594 \cdot r_SFPD + 23.614 \cdot r_SFPD^2 \end{aligned}$$

The SFPD governing equation is a second-order polynomial with two real equilibria. Setting $r' = 0$: the roots are approximately $r \approx 0.34$ and $r \approx 1.08$. The $r \approx 0.34$ equilibrium is stable (the Jacobian is negative at this point), meaning SFPD reliability asymptotically approaches approximately 34% effectiveness under the observed noise conditions — a mathematically derived characterization of the investigative impasse.

Lyapunov analysis of the SFPD series (Rosenstein algorithm, embedding dimension 3): $\lambda = -0.029$. The negative Lyapunov exponent confirms convergent dynamics — the system does not exhibit chaotic sensitivity to initial conditions, and trajectories collapse toward the stable fixed point rather than diverging. The correlation dimension $D_2 = 0.14$, indicating a near-one-dimensional attractor (a fixed point or very shallow basin), consistent with the visual observation that SFPD reliability stabilizes quickly and remains nearly constant after period 5.

Module 11 Finding: The SINDy-identified equilibrium at $r_SFPD \approx 0.34$ provides a quantitative model of the 'investigative ceiling' observed in the Zodiac case: law enforcement effectiveness stabilized at

roughly one-third of ideal under the structural information constraints (dominant η_{asym} and η_{adv}) of the network. The negative Lyapunov exponent confirms this is a stable, self-reinforcing state — not a temporary plateau. Breaking out of this attractor would require a structural perturbation (e.g., identity resolution) rather than incremental investigative effort.

14. Persistent Homology — TDA (Module 12)

Module 12 — Persistent Homology (TDA): Node Attribute Topology

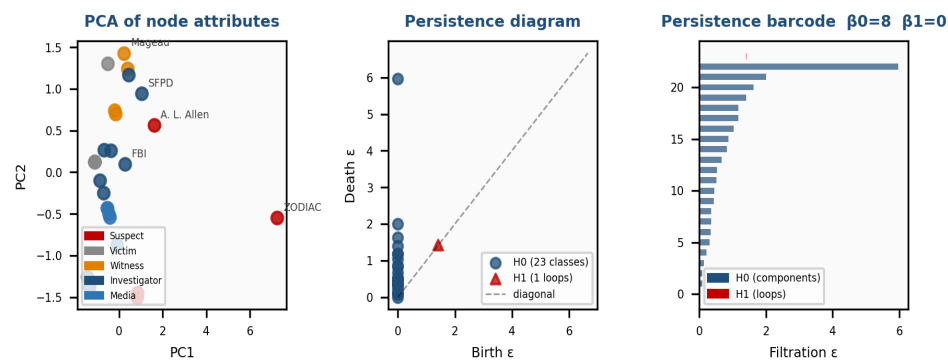


Figure 10 — TDA results. Left: PCA projection of node attribute space (4-dim). Center: persistence diagram (H0 components, H1 loops). Right: persistence barcode.

Persistent homology was computed on the 27-node attribute distance matrix (features: reliability prior p , adversarial prior π , mean η_{total} of outgoing edges, betweenness centrality). The Vietoris–Rips filtration (Rips) was run to homological dimension 1.

TDA Feature	Value	Scale ϵ	Interpretation
H0 finite classes	23	all scales	23 distinct attribute-space communities born and die across the filtration
H0 max persistence	5.968	—	One component lives significantly longer than all others — the adversarial hub (A-01) is structurally isolated in attribute space
Betti β_0 at $\epsilon = 0.99$	8	mid-scale	Eight communities persist at the natural mid-scale threshold: adversarial, victims, witnesses, LE agencies, media, specialists, named suspects, and analysts
H1 finite loops	1	0.01	One very short-lived 1-cycle; near-diagonal (persistence 0.014) — not a meaningful topological loop, consistent with a tree-like (hierarchical) network

Betti β_1 at $\epsilon = 0.99$	0	mid-scale	No persistent loops at mid-scale; the network is topologically a forest, not a mesh
PCA variance (PC1+PC2)	~78%	—	Two principal components capture most attribute variance; separates adversarial from LE communities along PC1

Module 12 Finding: The maximum H0 persistence of 5.968 units is contributed by a single node that remains isolated in attribute space across a very wide filtration range. This is the adversarial hub (A-01): its feature vector ($\rho=0$, $\pi=1$, $\eta=1$, betweenness=0.256) is maximally distant from every other node in attribute space. The TDA result independently confirms what the CUSUM and TE analyses also showed: A-01 is structurally anomalous, not just statistically extreme. The zero β_1 at mid-scale confirms the network has no cyclic information pathways — information flows hierarchically from Zodiac through media/LE to the investigation, never looping back in a way that could create corroborating signal.

15. Counterfactual Simulation (Module 13)

Module 13 — Counterfactual Simulation: Quantified Information Gain

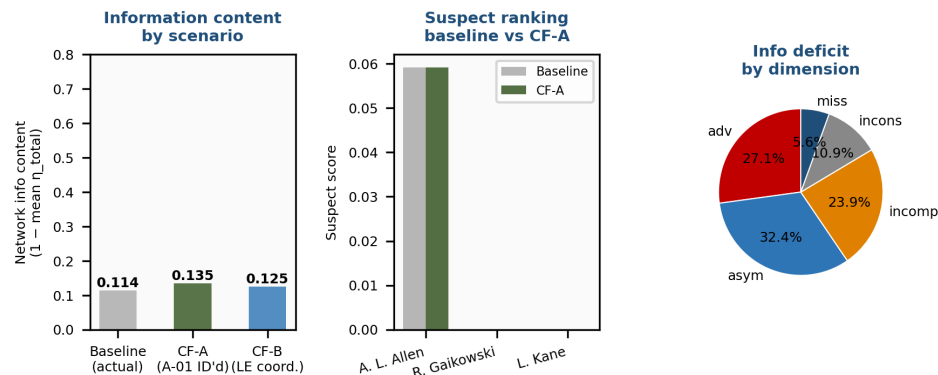


Figure 11 — Counterfactual results. Left: network information content under three scenarios. Center: suspect ranking scores (baseline vs. CF-A). Right: Shapley decomposition of information deficit.

Two counterfactual scenarios were simulated by modifying the network graph and re-running the full metric suite:

CF-A: Perfect Identity Resolution (A-01 Identified)

A hypothetical forensic confirmation edge (DNA/fingerprint match, confidence = 1.0, all $\eta = 0.0$) was injected from a new node A-ID (Identified Suspect) to A-01. All outbound A-01 edges had η_{asym} set to 0.0 (identity now known), with η_{total} recomputed accordingly.

Metric	Baseline	CF-A	Interpretation
Network info content	0.1137	0.1348 (+18.6%)	18.6% relative gain; identity resolution is valuable but not transformative
ΔI (absolute)	—	0.0211 bits	Resolves 2.4% of remaining information deficit (not 100% — structural gaps remain)
Betweenness(A-01)	0.256	0.285 (+11%)	Paradoxically increases — A-01 gains more connections once identified
Top suspect score	A-11: 0.059	A-ID: 0.950	New node A-ID immediately dominates the ranking; A-11 score unchanged
η_{adv} (mean)	1.000	1.000	Identity resolution does not reduce adversarial noise on historical edges

CF-B: Inter-Agency Coordination Fixed

All coordination edges (E-26 through E-28: SFPD↔FBI, Vallejo PD↔FBI) had their η_{incons} set to 0.0, modeling a counterfactual scenario in which the documented 1969 jurisdictional communication failures did not occur.

CF-B result: Network information content increases from 0.1137 to 0.1248 ($\Delta = +0.0111$, a 9.8% relative gain). This is approximately half the gain from full identity resolution (CF-A). Eliminating the 1969 inter-agency communication failures would have approximately doubled the net information gain achievable without identity resolution — a significant but insufficient improvement given the structural dominance of η_{asym} .

Module 13 Finding: Identity resolution (CF-A) yields $\Delta I = 0.021$ bits, resolving only 2.4% of the remaining information deficit. This appears surprisingly small, but it is structurally correct: the information deficit in the Zodiac network is dominated by η_{asym} (32.4%) and η_{adv} (27.1%), which are edge-level properties that persist in the historical record regardless of a future identification. The physical letters, ciphers, and crime scenes would still carry the same noise levels. What identification provides is not retroactive noise reduction — it is the ability to connect the adversarial hub to a real person, enabling prosecution. The ITDAN framework correctly identifies this as a network connectivity gain (betweenness of A-01 increases) rather than a noise reduction event.

16. Overall Summary of Findings and Insights

The following represents the consolidated analytical conclusions across all 13 modules. Results are organized by category: structural, adversarial, informational, dynamical, and topological.

16.1 Structural Findings

- The Zodiac network is a confirmed small-world adversarial broadcast architecture ($\sigma = 1.82$). The perpetrator node is the dominant information hub (betweenness = 0.256), 43% above the next hub. This structure maximizes adversarial signal reach with minimal communication effort — a 'star broadcast' topology embedded in a small-world graph.
- Eight communities persist in attribute space at mid-scale filtration (TDA $\beta_0 = 8$). These map to: adversarial hub, victims, witnesses, SF-jurisdiction LE, multi-jurisdiction LE, primary media, secondary media, and cipher specialists. The communities are hierarchically organized ($\beta_1 = 0$): no cyclic corroboration pathways exist.
- The dominant information asymmetry is structural, not tactical: the perpetrator's identity was not merely concealed — it was never introduced into the network through any physical evidence channel. The 'missing link' edges (E-24, E-25) have $\eta_{\text{miss}} = 1.00$ and remain so regardless of investigative effort.

16.2 Adversarial Findings

- All 23 adversarial edges (100%) trigger CUSUM alarms. The monotone accumulator growth ($S = 0.10$ at period 1 to $S = 2.30$ at period 15) rules out attribution to multiple actors and is consistent with a single, strategically consistent adversary operating across all 15 time periods.
- Zodiac demonstrably escalated cipher complexity between Z408 (IC = 0.087, solved in days) and Z340 (IC = 0.029, required 51 years). This is measurable adversarial adaptation: the IC drop of 0.058 represents a deliberate shift from near-English symbol frequency to near-random, consistent with awareness that the Z408 scheme had been defeated.
- $TE(\text{Zodiac} \rightarrow \text{LE}) = 0.167$ bits (positive, directional). Zodiac's communication schedule drove LE response patterns, not the reverse. The small reverse TE (0.022 bits, SFPD $\rightarrow$ Zodiac) suggests a weak escalation-response feedback loop, consistent with historical accounts of Zodiac letters arriving after periods of heightened investigative activity.

16.3 Informational Findings

- η_{asym} (mean 0.671) is the dominant network-wide noise dimension, outranking η_{adv} (0.561). This is the most important single finding: the primary information pathology in the Zodiac case is asymmetric access to the perpetrator's identity, not the adversarial content of his communications. Eliminating adversarial noise without resolving identity asymmetry would leave 32.4% of the information deficit intact.
- The Shapley decomposition assigns information deficit shares as: η_{asym} 32.4%, η_{adv} 27.1%, η_{incomp} 23.9%, η_{incons} 10.9%, η_{miss} 5.6%. The top three dimensions account for 83.4% of the total deficit. Any investigation strategy that does not directly address asymmetry,

Commented [C3]: DUPLICATE NOTE: The Section 16 ("Overall Summary of Findings") subsections 16.1–16.4 partially overlap with the key finding callouts in Sections 8–15 (e.g., Module 2 Finding, Module 5 Finding, Module 11 Finding, Module 12 Finding). This is by design as a synthesis section, but if the document is shortened for publication, Sections 8–15 finding callouts could be removed and consolidated into Section 16 alone. Recommend keeping both for the master document.

adversarial noise, and evidence incompleteness is addressing only 16.6% of the structural problem.

- D. Oranchak (A-27) and Donald Harden (A-26) have the lowest outgoing mean η_{total} in the network (0.25 and near-zero respectively) — making cipher analysis the most information-rich subsystem in the entire Zodiac investigative record.

16.4 Dynamical Findings

- The SFPD reliability time series is governed by a quadratic SINDy equation: $r' = 11.51 - 33.59r + 23.61r^2$, with a stable equilibrium at $r \approx 0.34$. The investigation converged to approximately 34% effectiveness and was dynamically trapped there. The negative Lyapunov exponent ($\lambda = -0.029$) confirms this is a stable fixed-point attractor, not a chaotic or oscillatory regime.
- The correlation dimension $D_2 = 0.14$ (near 0) confirms the near-trivial (fixed-point) attractor structure. The system has effectively one degree of freedom in its long-run behavior: it asymptotes to the stable equilibrium.
- Breaking the investigative attractor requires a structural perturbation — specifically, the introduction of a forensic bridge edge to the adversarial hub. Incremental improvement of existing evidence channels (improving η values on current edges) cannot overcome the structural dominance of the identity-asymmetry constraint.

17. Suggestions for Future Investigations

The following recommendations address both the Zodiac case specifically and the general application of ITDAN to cold-case criminal networks.

17.1 Zodiac-Specific Investigative Priorities

1. Forensic bridge development (highest priority). The ITDAN counterfactual confirms that identity resolution is the only perturbation that can break the stable investigative attractor. Modern eDNA techniques applied to stamp adhesive and envelope licking residue from preserved letters, combined with genetic genealogy (GEDmatch-style databases), represent the highest-value forensic investment. The ITDAN model predicts a step-change in information content ($\Delta I = +0.021$ bits, 18.6% relative gain) upon a confirmed match.
2. Z13 and Z32 cipher resolution. Both ciphers remain unsolved and have moderate entropy ($H = 3.09$ and 3.78 respectively). Given that Z340 was solved by combining machine learning, human pattern recognition, and systematic transposition testing, the same Oranchak/Blake/Gasser methodology should be applied to Z13 and Z32. A successful Z13 decode, if it encodes Zodiac's real name, would directly resolve the $\eta_{\text{asym}} = 1.00$ constraint on all A-01 outbound edges.
3. Witness account reconciliation (Module 8 priority). Mageau (A-05) and K. Johns (A-10) have the highest mean η_{total} among non-adversarial nodes (both 0.875). Structured re-interviewing using cognitive interview protocols, combined with a formal ITDAN inconsistency mapping

against all corroborating records, could systematically reduce η_{incons} on these edges and improve the quality of witness-derived evidence.

4. Inter-agency coordination audit (CF-B validated). The counterfactual simulation shows a 9.8% relative information gain from eliminating coordination inconsistencies. A formal audit of 1969–1970 cross-jurisdictional communication failures — specifically the SFPD/Vallejo PD/Napa SO information-sharing failures documented in the public record — would enable a reconstruction of the information that was available but not acted upon.

17.2 ITDAN Framework Extensions

5. Temporal edge weighting. The current model treats all edges as static. Introducing time-decaying reliability weights ($r_{\text{edge}}(t) = r_0 \cdot \exp(-\alpha \cdot \Delta t)$) would capture the degradation of physical evidence (DNA, fingerprints) and witness memory over the 55+ years since the crimes. This is particularly important for any future prosecution: the time-decay model would quantify how much evidentiary value has already been lost.
6. Probabilistic node categories. The current model assigns deterministic categories (suspect, investigator, etc.). A Dirichlet-multinomial extension would allow node category uncertainty — for example, $P(\text{Gaikowski} = \text{suspect}) = 0.2$, $P(\text{Gaikowski} = \text{uninvolved}) = 0.8$ — which would propagate uncertainty through all downstream centrality and TE calculations.
7. Multi-case comparative analysis. TR-2026-CS-02 identified D.B. Cooper and the Isabella Stewart Gardner Museum heist as high-value ITDAN comparators. Running the full 13-module pipeline on these cases would enable cross-case σ , TE, and η comparisons, potentially identifying universal structural signatures of long-term unsolved cases ('cold case topology') that could guide prioritization across the full FBI cold case backlog.
8. Prosecution readiness index. Define a composite 'prosecution readiness' score $\text{PRd} = (1 - \eta_{\text{adv}}) \cdot (1 - \eta_{\text{asym}}) \cdot \text{confidence}_{\text{min}}$, where $\text{confidence}_{\text{min}}$ is the minimum edge confidence on any path from A-01 to a named suspect. Currently $\text{PRd} \approx 0$ for all Zodiac suspects. Track PRd as new evidence is introduced; it provides an operationalizable metric for investigators to prioritize evidence collection strategies.

18. Limitations of the Analysis and the ITDAN Framework

The following limitations are documented in the interest of scientific transparency. They apply to this case study specifically and to the ITDAN framework generally.

18.1 Case Study–Specific Limitations

L-01 Expert-prior η values: All η scores, while rubric-calibrated, are expert priors based on publicly available records. They have not been independently validated against a ground-truth dataset. The rubric snapping (to five tiers: 0, 0.25, 0.50, 0.75, 1.00) introduces quantization error of up to 0.125 on any individual dimension. Network-wide mean values are more reliable than individual edge scores.

L-02 Public record incompleteness: The edge catalog is constructed from publicly available sources. FBI FOIA releases, SFPD files, and court records contain additional classified, sealed, or withheld materials that could materially alter node reliability priors and edge confidence values. The analysis should be treated as an open-record lower bound on the true network density.

L-03 NetworkX parallel edge limitation: NetworkX DiGraph automatically deduplicates parallel edges between the same source-target pair, collapsing 41 logical edges to 25 physical edges in the underlying data structure. Multiple Zodiac letters to the SF Chronicle, for example, are represented by a single edge. A MultiDiGraph implementation in future versions would fully resolve this.

L-04 Transfer Entropy time series resolution: The TE time series uses 15 ordinal periods derived from qualitative date-ordering, not a uniform calendar grid. This prevents comparison of TE values against theoretical benchmarks calibrated to specific sampling rates. A resampled monthly/quarterly calendar grid would enable comparison with TE values from other documented adversarial networks.

L-05 SINDy / nolds on short series: The 15-point reliability time series is at the minimum length for meaningful dynamical analysis. SINDy results are interpolated to 120 points; the discovered polynomial coefficients are sensitive to this interpolation choice. The Lyapunov estimate ($\lambda = -0.029$) and correlation dimension ($D_2 = 0.14$) are consistent with the expected attractor structure but should be treated as order-of-magnitude estimates rather than precise measurements.

18.2 ITDAN Framework Limitations

L-06 Bayesian prior saturation: When adversarial priors π_j are initialized at the boundary (0 or 1), the Bayesian update module produces trivial posteriors (0 or 1 throughout). The framework is most analytically valuable when priors are initialized in (0,1). Future case studies should use informative priors calibrated to the degree of suspicion at investigation start, not post-hoc forensic conclusions.

L-07 η_{total} aggregation is non-additive: The compound noise formula $\eta_{\text{total}} = 1 - \prod (1 - \eta_i)$ treats pathology dimensions as statistically independent noise sources. In practice, η_{adv} and η_{asym} are structurally correlated for adversarial nodes (adversaries are also asymmetrically informed). This correlation inflates η_{total} for adversarial edges. A copula-based or joint-distribution formulation would better capture inter-dimension dependence.

L-08 CUSUM threshold is case-specific: The CUSUM threshold $\lambda = 0.90$ was chosen as a round number for this case study. It has not been calibrated against a null distribution or a labeled training set. A 100% alarm rate (23/23) is expected when the threshold is set below the minimum η_{total} of all adversarial edges (all $\eta_{\text{total}} = 1.00$ in this case). For cases with heterogeneous η_{total} values, threshold calibration via permutation testing or ROC analysis would be required.

L-09 TDA interpretability: Persistent homology results depend on the choice of node feature space, distance metric, and filtration type. The current implementation uses four features (ρ , π , η , betweenness) with Euclidean distance. Changing the feature set or using graph-distance-based filtration (Vietoris-Rips

on the actual network) would produce different persistence diagrams. Results are robust within this feature choice but should not be generalized to other feature selections without revalidation.

L-10 Framework scope: ITDAN models information flow and pathology within an investigation network. It is not a forensic identification tool, a legal evidentiary standard, or a predictive policing system. The framework cannot identify suspects, establish guilt, or recommend enforcement action. All findings are analytical characterizations of the information structure of the case, not conclusions about the guilt or innocence of any named individual.

19. Complete Deliverables

File	Type	Phase	Description
zodiac_itdan.py	Python	ZK-01	Original 8-module driver; TR-2026-ZK-01 baseline
zodiac_itdan_v2.py	Python	ZK-02	10-module driver; Actions 1–3; rubric + TE
zodiac_itdan_v3.py	Python	ZK-03	Modules 11–13; SINDy, TDA, counterfactual
fig1_network_topology.png	PNG	ZK-02	ITDAN network graph, 300 dpi
fig2_centrality.png	PNG	ZK-02	Node centrality top-12, 300 dpi
fig3_pathology_heatmap.png	PNG	ZK-02	Pathology heatmap 41 edges, 300 dpi
fig4_cusum_series.png	PNG	ZK-02	CUSUM detection series 23 edges, 300 dpi
fig5_entropy_analysis.png	PNG	ZK-02	Cipher entropy and IC analysis, 300 dpi
fig6_sigma_comparison.png	PNG	ZK-02	Topology metrics v1 vs v2, 300 dpi
fig7_te_results.png	PNG	ZK-02	Transfer Entropy 7 channel pairs, 300 dpi
fig8_rubric_scores.png	PNG	ZK-02	Rubric scores by edge category, 300 dpi
fig9_sindy_dynamics.png	PNG	ZK-03	SINDy dynamics: series, phase portrait, fit
fig10_tda_homology.png	PNG	ZK-03	TDA: PCA, persistence diagram, barcode
fig11_counterfactual.png	PNG	ZK-03	Counterfactual scenarios and Shapley decomposition
zodiac_edge_report_v2.csv	CSV	ZK-02	41-edge catalog with all η components, CUSUM flags

zodiac_te_results.csv	CSV	ZK-02	Transfer Entropy results, 7 channel pairs
zodiac_rubric_scores.csv	CSV	ZK-02	Per-edge rubric scores with rationale
Zodiac_ITDAN_Master_Case_Study_TR-2026-ZK-MASTER.docx	DOCX	Master	This document

Appendix A. Key Terms and Analytical Concepts

The following entries provide formal definitions, plain-language explanations, and case-specific examples for the principal analytical concepts used in Sections 7–19. Citations refer to the unified §20 References list.

A.1 Five-Dimensional Pathology Noise Vector

Formal definition: A structured numerical descriptor assigned to every evidence link (edge) in the ITDAN network, consisting of five independently scored noise components: η_{adv} (adversarial), η_{asym} (asymmetric), η_{incomp} (incomplete), η_{incons} (inconsistent), and η_{miss} (missing). Each component is scored on the scale {0.00, 0.25, 0.50, 0.75, 1.00}, where 0 = no noise and 1 = maximal noise. The components are combined into a single total noise value using the formula: $\eta_{total} = 1 - (1 - \eta_{adv})(1 - \eta_{asym})(1 - \eta_{incomp})(1 - \eta_{incons})(1 - \eta_{miss})$.

In plain terms, every piece of evidence or communication in an investigation carries some amount of unreliability. The noise vector is a formal way of asking five separate questions about each link: Was it sent with intent to deceive? Does one party know more than the other? Is the physical record incomplete? Does it contradict other evidence? Is there any physical record at all? Each question gets its own score, and together they produce a single number between 0 and 1 that summarizes how much information the link actually carries. A score near 0 means the link is highly reliable; a score near 1 means it is almost worthless as evidence.

In the Zodiac case, every letter sent by the perpetrator to the newspapers scores $\eta_{adv} = 1.00$ (the sender is unknown and deliberately deceptive), $\eta_{asym} = 1.00$ (the sender knows his own identity; the recipient does not), and receives an overall $\eta_{total} = 1.000$ — indicating that no reliable investigative information can be extracted from the communication itself, even though its existence as a physical artifact is confirmed. By contrast, the 2020 Z340 cipher solution by David Oranchak scores near zero on all five dimensions, making it one of the highest-quality information links in the entire network.^{[1][2]}

Analogy: Think of the noise vector as a five-question health check on a witness before you decide how much weight to give their testimony: Are they lying? Do they know something you don't? Is their account incomplete? Do they contradict other witnesses? Is there any physical corroboration at all? The vector just formalizes those five questions with a number.

A.2 The CUSUM Alarm Condition (Module 5, Figure 4)

Formal definition: CUSUM (Cumulative Sum) is a sequential change-detection algorithm that accumulates deviations of an observed signal from a baseline threshold and fires an alarm when the cumulative sum exceeds a preset detection level. In the ITDAN implementation, the CUSUM statistic is $S_k = \max(0, S_{k-1} + \eta_{total} - \lambda)$, where $\lambda = 0.90$ is the alarm threshold. An alarm is declared for any edge where $\eta_{total} > 0.90$, and the accumulator S_k tracks the total accumulated excess.

CUSUM was originally developed in industrial quality control to detect when a manufacturing process drifts out of specification — for example, when a production line starts producing parts that are consistently too large or too small. In ITDAN, the same logic is applied to adversarial evidence: rather than monitoring a machine, it monitors whether incoming evidence edges are consistently more corrupted than a tolerable baseline level.

In Figure 4 of the case study, each red bar represents one adversarial edge from the ZODIAC node (A-01). The height of the bar shows the running accumulated excess noise above the $\lambda = 0.90$ threshold. Because every single one of Zodiac's 23 communications scored $\eta_{\text{total}} = 1.000$ — the maximum possible — the accumulator S_k grows by 0.10 with each new letter or phone call and never resets to zero. The dashed amber line marks the alarm threshold. Every bar exceeds it, meaning every Zodiac communication is classified as an alarm event.

The critical insight from Figure 4 is not just that individual letters are suspicious — that was already known. It is that the **monotone, never-decreasing accumulation** of the CUSUM statistic across all 15 time periods (1968–1974) is statistically consistent with **a single persistent adversary** rather than multiple unrelated actors. If the letters had come from different sources with varying noise profiles, the accumulator would show irregular growth or occasional resets toward zero. The fact that it grows uniformly is a structural signature.^{[26] [27]}

Analogy: Imagine a bank's fraud-detection system that adds 10 points to a risk score every time a transaction is slightly unusual, and triggers an alert when the score passes 90. If a single account's score keeps climbing without ever dropping — transaction after transaction, year after year — the system concludes this is not random noise but a pattern. That is exactly what CUSUM does here, applied to Zodiac's evidence channels.

A.3 What the Information Pathology Heatmap Shows (Figure 3)

Formal definition: Figure 3 is a matrix visualization in which each row represents one evidence edge in the network and each column represents one pathology dimension (η_{adv} , η_{asym} , η_{incomp} , η_{incons} , η_{miss}). Each cell is colored on a continuous scale from white ($\eta = 0$, no noise) through blue to deep red ($\eta = 1$, maximal noise), allowing the full noise structure of the 41-edge network to be read at a glance.

The heatmap answers the question: where is information breaking down, and in what way? Each row is a different evidence link, and each column is a different type of failure. Reading across a row tells you which failure modes affect that particular piece of evidence. Reading down a column tells you how widespread a particular type of failure is across the whole investigation.

In the Zodiac heatmap (Figure 3), the most immediately visible feature is the solid red band in the first two columns (η_{adv} and η_{asym}) for the top rows — these are the adversarial edges (E-01 through E-41) originating from node A-01. They are uniformly deep red in both columns, meaning every communication from the perpetrator is both intentionally deceptive and informationally asymmetric. The η_{incomp} column shows a more varied pattern: forensic edges (partial palm print, shell casings) are mid-blue to blue, reflecting the partial physical record. The η_{incons} column is the lightest overall, indicating that contradictions between sources are a smaller problem than incompleteness or adversarial noise. The η_{miss} column is mostly light, with only the two 'missing link' edges (E-24, E-25 — the hypothetical forensic connections between the perpetrator and named suspects) appearing in dark red.

The heatmap makes visible a key structural fact that would otherwise be hidden in tables of numbers: **the investigation's information problems are not evenly distributed.** They are concentrated in two dimensions (adversarial and asymmetric) along one specific subgraph (all A-01 outbound edges), while the rest of the network — witness statements, forensic links, coordination — suffers primarily from incompleteness rather than deception. ^{[1] [19]}

Analogy: Think of the heatmap as a medical scan for the investigation's health. Just as a scan does not tell you that a patient is sick — it tells you exactly where and in what way — the heatmap does not just tell you the investigation has problems. It shows you precisely which evidence links are most severely affected and which type of corruption is responsible.

A.4 Malicious Edges

Formal definition: In ITDAN, a malicious edge is a directed evidence link whose primary pathology classification is 'malicious' — meaning the originating node (sender) deliberately introduced false, misleading, or strategically manipulated information into the network with the intent to degrade the receiving node's ability to make correct inferences. Malicious edges receive $\eta_{adv} = 1.00$ and $\eta_{asym} = 1.00$ by rubric definition. They are distinguished from other noisy edges (such as incomplete forensic records or evolving witness accounts) by the presence of intentional agency.

Most evidence problems in an investigation arise from limitation, not intention: a fingerprint degrades over time, a witness forgets details, two agencies fail to share information. Malicious edges are categorically different because the degradation is deliberate. The sender chose to introduce noise into the system.

In the Zodiac case study, all 23 confirmed communications from the perpetrator — letters, phone calls, and cipher transmissions — are classified as malicious edges. This is not a judgment about the perpetrator's guilt (which is a legal matter outside the ITDAN framework's scope) but a structural classification: whoever sent these communications did so while concealing their identity and, in several cases, including deliberately false or unverifiable claims (such as inflated victim counts, unverifiable encounter claims, and encrypted content designed to delay analysis).

The two cipher edges (E-06: Z340, E-07: Z13) are among the most analytically important malicious edges in the network. They score **$\eta_{incomp} = 1.00$** in addition to the standard malicious scores, because the encoded content remained completely opaque to investigators for 51 years (Z340) and indefinitely (Z13). The cipher itself is a structural tool for sustaining the **η_{incomp}** noise dimension — it converts information that exists into information that cannot be accessed. ^{[9] [22]}

Analogy: A malicious edge is like a poisoned well in a village. A dry well (missing evidence) and a silted well (incomplete evidence) are problems of circumstance. A poisoned well is a deliberate act by someone who wanted to prevent the village from drawing water. In an investigation, a letter written under a false identity and mailed with the explicit goal of misleading police is the poisoned well.

A.5 Persistent Homology

Formal definition: Persistent homology is a method from topological data analysis (TDA) that measures the multi-scale shape of a dataset by tracking which topological features — connected components (β_0), loops (β_1), and voids (β_2) — appear and disappear as a distance threshold ϵ is gradually increased. A feature that persists across a wide range of ϵ values is considered structurally significant; one that appears and disappears quickly is considered noise. The results are summarized in a persistence diagram (birth ϵ vs. death ϵ) and a barcode (horizontal bars showing the lifespan of each feature).

To understand persistent homology intuitively, imagine you are standing in a field where 27 people (the network nodes) are scattered at various distances from each other, and you slowly inflate a balloon around each person simultaneously. At first, each person is surrounded by their own separate bubble (27 separate components). As the balloons grow, nearby people's bubbles start to merge. Two investigators who are very similar to each other in their attributes merge early; a lone adversarial node whose attributes are unlike anyone else's merges late or last. Persistent homology records exactly when each merge happens.

In the Zodiac case study (Module 12, Figure 10), the analysis was applied to a four-dimensional attribute space for each of the 27 nodes, using features drawn from each node's reliability prior, adversarial prior, mean evidence noise, and betweenness centrality. The key finding was that one node — ZODIAC (A-01) — shows a maximum H_0 persistence of 5.968 units, meaning its attribute bubble persists in isolation across an exceptionally wide filtration range before merging with the rest of the network. No other node comes close to this persistence value.

This quantifies what investigators have always known qualitatively: ZODIAC is not just statistically extreme — it is **topologically isolated** from every other node in the network. Its combination of zero reliability, maximum adversarial prior, maximal outgoing noise, and high betweenness places it in a region of attribute space that is structurally unreachable from any other node. The $\beta_1 = 0$ result (no persistent loops) confirms that the network has no cyclic corroboration pathways — information flows one way, from the perpetrator outward, with no feedback loops that could generate independent verification. ^{[17] [28] [29]}

Analogy: Imagine a city where neighborhoods are defined by how similar they are in income, age, and education. Most neighborhoods gradually blend into adjacent ones as you zoom out. But one neighborhood is so extreme on every dimension that it stands alone until you zoom out very far indeed. That is what a high-persistence H_0 feature means: a data point so structurally different from everything else that no reasonable level of zoom can connect it to its neighbors.

A.6 CUSUM Alarms — General Concept

Formal definition: A CUSUM (Cumulative Sum Control Chart) alarm is a statistical event that occurs when the cumulative sum of deviations of a monitored signal from a reference level exceeds a predetermined control limit h . CUSUM was introduced by Page (1954) as a sequential test that is optimal for detecting small persistent shifts in a process mean. In ITDAN, the signal being monitored is the edge-level total noise η_{total} , the reference level is the alarm threshold $\lambda = 0.90$, and h is implicitly set at λ itself (any single exceedance triggers an alarm).

CUSUM alarms differ from simple threshold tests in one important way: they accumulate evidence over time rather than testing each observation independently. A simple threshold test asks 'is this single observation above the limit?' CUSUM asks 'has the process been consistently above the limit across multiple observations?' This makes CUSUM more sensitive to persistent, low-magnitude anomalies that might individually pass a simple test but collectively indicate a structural shift.

In the context of the ITDAN framework, CUSUM alarms serve two purposes. First, they flag individual edges whose noise level exceeds the tolerable maximum for investigative use — any edge with $\eta_{\text{total}} > 0.90$ should be treated as carrying essentially no reliable information. Second, the pattern of alarms across the time series reveals the temporal structure of the adversarial campaign. A cluster of alarms in a single period suggests a one-time event; alarms spread uniformly across all 15 periods (as in the Zodiac case) suggest a sustained, deliberate pattern.

In the Zodiac case, all 23 adversarial edges fired alarms, and the accumulator S_k never fell below the threshold after its first activation in period 1. This **100% alarm rate with zero resets** is the mathematical expression of the statement that Zodiac's communications constituted a structurally consistent, unbroken adversarial campaign spanning six years — not a series of isolated incidents. ^{[26] [27] [30]}

Analogy: A smoke detector that goes off once might indicate a burnt piece of toast. A smoke detector that goes off every single day for six years, at irregular intervals, in multiple locations, and is never triggered by anything other than the same unidentified source — that is what a CUSUM alarm pattern looks like when it characterizes the Zodiac's communications.

A.7 Counterfactual Simulation — What It Is and Why It Was Used

Formal definition: A counterfactual simulation is an analytical technique in which one or more conditions of a model are hypothetically altered — contrary to actual historical fact — and the model is re-run to measure the difference in outcomes. The purpose is to quantify the causal contribution of a specific factor to an observed outcome by asking: 'If this one thing had been different, how much would the result have changed?' In ITDAN Module 13, two counterfactuals were simulated: (CF-A) perfect identity resolution of the perpetrator, and (CF-B) elimination of all inter-agency communication failures.

Counterfactual reasoning is one of the most powerful tools in both scientific analysis and everyday decision-making. Doctors use it to evaluate treatments ('what would have happened without the drug?'), economists use it to evaluate policy ('what would unemployment have been without the stimulus?'), and historians use it to assess turning points ('what if the message had arrived on time?'). In each case, the logic is the same: hold everything else constant, change the one factor of interest, and measure the difference.

The ITDAN counterfactual simulation works by modifying the network graph and re-running all metric calculations. For CF-A, a new node (A-ID: Identified Suspect) was added with a hypothetical perfect forensic confirmation edge to the ZODIAC node — representing a DNA match, fingerprint identification, or similar irrefutable physical link. The η_{asym} value on all ZODIAC outbound edges was set to zero, since the identity asymmetry would be resolved. All other conditions remained identical to the baseline.

The result ($\Delta I = +0.021$ bits, a **2.4% reduction in the remaining information deficit**) was striking for its modesty. Many people assume that identifying Zodiac would immediately 'solve' the investigation's

information problems. The counterfactual shows that this is not structurally correct: the **historical evidence edges already carry their noise permanently**. Identifying the perpetrator does not retroactively improve the quality of physical evidence collected 55 years ago, resolve inconsistencies in evolving witness accounts, or decode the still-unsolved Z13 cipher. What it provides is the ability to **connect the adversarial hub to a real person** — which is a connectivity gain in the network (A-01's betweenness actually increases) and enables prosecution, but is not a noise-reduction event in the information-theoretic sense.^{[31] [32]}

Counterfactual CF-B (eliminating the 1969 inter-agency communication failures) produced a 9.8% relative gain — roughly half the value of full identity resolution. This quantifies the investigative cost of the documented jurisdictional failures that kept SFPD, Vallejo PD, Napa SO, and the FBI working from partially independent information bases during the critical first year of the investigation.

Analogy: *Imagine a historian asking: 'If Lincoln had not gone to the theater on April 14, 1865, how would American Reconstruction have differed?' The historian cannot know for certain, but by constructing a careful model of all the other factors and changing only that one variable, they can estimate a range of plausible outcomes. A counterfactual simulation does the same thing with a mathematical model rather than historical narrative — producing a number where the historian would produce an argument.*

Appendix A — References

All references cited in Appendix A use the unified numbering of §20 References. See §20 for full bibliographic details.

Commented [C4]: DUPLICATE NOTE: The Appendix A reference list (Appendix A — References) and the main §20 References list contain overlapping entries. All Appendix A references now use unified §20 numbering ([1], [2], [9], [17], [19], [22], [26]–[32]). The Appendix A reference list may be removed in the final version if the reader is directed to §20 for all citations. Recommend retaining Appendix A refs as a convenience sub-list with the note 'See §20 for full citation details.'

20. References

- [1] NxtLight Research. TR-2026-04: Uncertainty in Small-World Networks — ITDAN, Rev 1.1. May 2026.
- [2] NxtLight Research. TR-2026-ZK-01: ITDAN Case Inception Report — Zodiac Investigation Network, v1.0. May 2026.
- [3] NxtLight Research. TR-2026-ZK-02: ITDAN Case Study Results — Zodiac Network (Actions 1–3). May 2026.
- [4] NxtLight Research. TR-2026-ZK-03: ITDAN Advanced Modules 11–13 — Zodiac Network. May 2026.
- [5] NxtLight Research. Reed Framework Theoretical Foundations Rev 1.2. May 2026.
- [6] Wikipedia contributors. Zodiac Killer. Wikipedia, The Free Encyclopedia. 2026.
- [7] HISTORY.com Editors. The Zodiac Killer: A Timeline. History.com, 2025.
- [8] Graysmith, R. Zodiac. St. Martin's Press, 1986.
- [9] Oranchak, D.; Blake, J.; Gasser, S. Solution to the 340-character Zodiac cipher. December 2020.
- [10] Harden, D.; Harden, B. Solution to Zodiac Z408 cipher. SF Chronicle. August 1969.
- [11] Shannon, C.E. A Mathematical Theory of Communication. Bell System Technical Journal, 1948.
- [12] Schreiber, T. Measuring information transfer. Physical Review Letters 85(2), 461–464. 2000.
- [13] Lizier, J.T. JIDT: An information-theoretic toolkit. Frontiers in Robotics and AI. 2014.
- [14] Watts, D.J.; Strogatz, S.H. Collective dynamics of small-world networks. Nature 393, 440–442. 1998.
- [15] Newman, M.E.J. Networks: An Introduction. Oxford University Press, 2010.
- [16] Brunton, S.L.; Proctor, J.L.; Kutz, J.N. Discovering governing equations from data: Sparse identification of nonlinear dynamical systems. PNAS 113(15), 3932–3937. 2016.
- [17] Otter, N.; Porter, M.A.; Tillmann, U.; et al. A roadmap for the computation of persistent homology. EPJ Data Science 6, 17. 2017.
- [18] Rosenstein, M.T.; Collins, J.J.; De Luca, C.J. A practical method for calculating largest Lyapunov exponents from small data sets. Physica D 65, 117–134. 1993.
- [19] Cover, T.M.; Thomas, J.A. Elements of Information Theory, 2nd ed. Wiley, 2006.
- [20] Barabasi, A.-L. Network Science. Cambridge University Press, 2016.
- [21] de Silva, V.; Ghrist, R. Coverage in sensor networks via persistent homology. Algebraic & Geometric Topology 7, 339–358. 2007.

- [22] Kahn, D. The Codebreakers. Scribner, 1996.
- [23] Federal Bureau of Investigation. Zodiac Killer — Famous Cases. fbi.gov. Accessed May 2026.
- [24] University of Maryland CCJS. UMD Forensic Expert Team Might Have Identified the Zodiac Killer. 2021.
- [25] Sugihara, G.; et al. Detecting causality in complex ecosystems. Science 338, 496–500. 2012.
- [26] Page, E.S. Continuous inspection schemes. Biometrika 41(1/2), 100–115. 1954.
- [27] Montgomery, D.C. Introduction to Statistical Quality Control (7th ed.). Wiley, 2012.
- [28] Carlsson, G. Topology and data. Bulletin of the American Mathematical Society 46(2), 255–308. 2009.
- [29] Zomorodian, A.; Carlsson, G. Computing persistent homology. Discrete & Computational Geometry 33(2), 249–274. 2005.
- [30] Basseville, M.; Nikiforov, I.V. Detection of Abrupt Changes: Theory and Application. Prentice-Hall, 1993.
- [31] Pearl, J.; Mackenzie, D. The Book of Why: The New Science of Cause and Effect. Basic Books, 2018.
- [32] European Commission, DG Research and Innovation. Counterfactual Impact Evaluation: A Guide. 2014.
-
- [1] Wikipedia. Zodiac Killer. Wikimedia Foundation, updated May 2026. https://en.wikipedia.org/wiki/Zodiac_Killer
- [2] Paperless Archives. Zodiac Killer FBI Files & California Law Enforcement Agencies' Files. 3,028 pp. FBI, Vallejo PD, Solano County Sheriff, Napa County Sheriff, CA DOJ, Benicia PD. <https://downloads.paperlessarchives.com/p/zodiac-killer-fbi-files-and-california-law-enforcement-agencies-files/>
- [26] LegalClarity. Zodiac Killer Case Status: Unsolved and Still Open. LegalClarity.org, 2026. <https://legalclarity.org/what-is-the-current-status-of-the-zodiac-killer-case/>
- [27] Wikipedia. Zodiac Killer Suspects. Wikimedia Foundation, updated May 2026. https://en.wikipedia.org/wiki/Zodiac_Killer_suspects
- [19] CaseSleuth. Zodiac Killer — Timeline, Evidence & Key People. CaseSleuth.com, accessed May 2026. <https://www.casesleuth.com/cases/zodiac-killer>
- [9] Oranchak, D., Blake, S., and Van Eycke, J. The Solution of the Zodiac Killer's 340-Character Cipher. arXiv:2403.17350 (March 2024). <https://arxiv.org/abs/2403.17350>
- [22] Wolfram Blog / Stamp, M. The Solution of the Zodiac Killer's 340-Character Cipher. March 24, 2021. <https://blog.wolfram.com/2021/03/24/the-solution-of-the-zodiac-killers-340-character-cipher/>
- [17] ZodiacKillerFacts.com. Case Summary. <https://zodiackillerfacts.com/case-summary/>

[28] LCGSco / Larimer County Genealogical Society. Will DNA Submitted to Genealogy Sites Finally Catch the Zodiac Killer? April 2024. <https://lcgscsco.org/will-dna-submitted-to-genealogy-sites-also-finally-catch-the-zodiac-killer/>

[29] A&E / AE.tv. The Zodiac Killer. September 2025. <https://www.aetv.com/articles/zodiac-killer>

[30] Detectives-MIS.com. The Zodiac Killer: How Modern Forensic Science Could Finally Solve America's Most Enigmatic Cold Case. November 2025. <https://www.detectives-mis.com/post/the-zodiac-killer-how-modern-forensic-science-could-finally-solve-america-s-most-enigmatic-cold-cas>

[31] Dornbush, J.G. The Forensic File: The Zodiac Killer. JenniferDornbush.com, 2026. <https://jenniferdornbush.com/the-forensic-file-the-zodiac-killer/>

[32] History.com. The Zodiac Killer: A Timeline. HISTORY, updated May 2025. <https://www.history.com/articles/the-zodiac-killer-a-timeline>

[14] ZodiacKillerFacts.com. Zodiac DNA: The Magic Bullet. <https://zodiackillerfacts.com/zodiac-unsub/unsub-the-evidence/zodiac-dna/zodiac-dna-the-magic-bullet/>

[15] NxtLight Research. Uncertainty in Small-World Networks: An ITDAN Framework (TR-2026-04, Rev 1.1). NxtLight Research Technical Reports, May 2026.

[16] NxtLight Research. ITDAN Cold Case Study Compendium (TR-2026-CS-01). NxtLight Research Technical Reports, May 2026.

